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Barlow Twins: Self-Supervised Learning via Redundancy Reduction

Jure Zbontar [*1] **Li Jing** [*1] **Ishan Misra** [1] **Yann LeCun** [1 2] **Stéphane Deny** [1]

Abstract

Self-supervised learning (SSL) is rapidly closing the gap with supervised methods on large computer vision benchmarks. A successful approach to SSL is to learn embeddings which are invariant to distortions of the input sample. However, a recurring issue with this approach is the existence of trivial constant solutions. Most current methods avoid such solutions by careful implementation details. We propose an objective function that naturally avoids collapse by measuring the cross-correlation matrix between the outputs of two identical networks fed with distorted versions of a sample, and making it as close to the identity matrix as possible. This causes the embedding vectors of distorted versions of a sample to be similar, while minimizing the redundancy between the components of these vectors. The method is called BARLOW TWINS, owing to neuroscientist H. Barlow's *redundancy-reduction principle* applied to a pair of identical networks. BARLOW TWINS does not require large batches nor asymmetry between the network twins such as a predictor network, gradient stopping, or a moving average on the weight updates. Intriguingly it benefits from very high-dimensional output vectors. BARLOW TWINS outperforms previous methods on ImageNet for semi-supervised classification in the low-data regime, and is on par with current state of the art for ImageNet classification with a linear classifier head, and for transfer tasks of classification and object detection.[1]

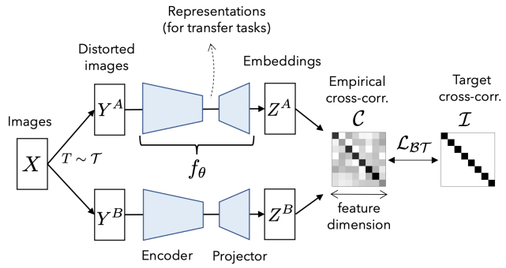


Figure 1. BARLOW TWINS's objective function measures the cross-correlation matrix between the embeddings of two identical networks fed with distorted versions of a batch of samples, and tries to make this matrix close to the identity. This causes the embedding vectors of distorted versions of a sample to be similar, while minimizing the redundancy between the components of these vectors. BARLOW TWINS is competitive with state-of-the-art methods for self-supervised learning while being conceptually simpler, naturally avoiding trivial constant (i.e. collapsed) embeddings, and being robust to the training batch size.

1. Introduction

Self-supervised learning aims to learn useful representations of the input data without relying on human annotations. Recent advances in self-supervised learning for visual data (Caron et al., 2020; Chen et al., 2020a; Grill et al., 2020; He et al., 2019; Misra & van der Maaten, 2019) show that it is possible to learn self-supervised representations that are competitive with supervised representations. A common underlying theme that unites these methods is that they all aim to learn representations that are invariant under different distortions (also referred to as 'data augmentations'). This

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[1]Code and pre-trained models (in PyTorch) are available at https://github.com/facebookresearch/barlowtwins

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Barlow Twins：通过冗余消减进行自监督学习

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摘要

自监督学习（SSL）正在大型计算机视觉基准上迅速缩小与监督方法的差距。一种成功思路是学习对输入样本扰动保持不变的嵌入。然而，这种方法反复面临平凡常量解的问题。当前多数方法依靠精心安排的实现细节来避免这种解。我们提出一种能自然避免坍塌的目标函数：让两个相同网络分别处理样本的不同扰动版本，计算输出之间的交叉相关矩阵，并使其尽可能接近单位矩阵。这样，同一样本不同扰动版本的嵌入向量会彼此相似，同时向量各分量之间的冗余被尽量减少。该方法称为 BARLOW TWINS，名称源于将神经科学家 H. Barlow 的冗余消减原则用于一对相同网络。BARLOW TWINS 无需大批量，也无需在双网络之间引入预测器、停止梯度或权重更新的移动平均等非对称机制。有趣的是，它受益于维度很高的输出向量。在少量标注数据的 ImageNet 半监督分类中，它优于以往方法；在使用线性分类头的 ImageNet 分类，以及分类和目标检测迁移任务上，其表现与当前最优方法相当。[1]

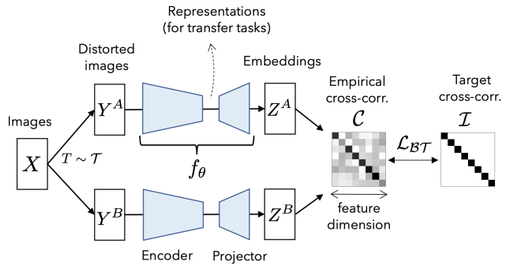


图1. BARLOW TWINS 的目标函数衡量两个相同网络的嵌入之间的交叉相关矩阵；网络接收同一批样本的不同扰动版本，并使该矩阵接近单位矩阵。这使同一样本不同扰动版本的嵌入向量相似，同时尽量减少向量各分量之间的冗余。BARLOW TWINS 的自监督学习性能可媲美当前最优方法，同时概念更简单，能自然避免平凡的常量嵌入（即坍塌），且对训练批量大小具有鲁棒性。

1. 引言

自监督学习旨在不依赖人工标注的情况下，学习输入数据的有用表征。视觉数据自监督学习的近期进展（Caron et al., 2020; Chen et al., 2020a; Grill et al., 2020; He et al., 2019; Misra & van der Maaten, 2019）表明，自监督表征可以与监督表征相媲美。这些方法的共同主题，是学习对不同扰动（也称为“数据增强”）保持不变的表征。这

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第 38th 届国际机器学习会议论文集，PMLR 139，2021。版权归作者所有，2021。[1]代码与预训练模型（PyTorch）见 https://github.com/facebookresearch/barlowtwins

is typically achieved by maximizing similarity of representations obtained from different distorted versions of a sample using a variant of Siamese networks (Hadsell et al., 2006). As there are trivial solutions to this problem, like a constant representation, these methods rely on different mechanisms to learn useful representations.

Contrastive methods like SIMCLR (Chen et al., 2020a) define 'positive' and 'negative' sample pairs which are treated differently in the loss function. Additionally, they can also use asymmetric learning updates wherein momentum encoders (He et al., 2019) are updated separately from the main network. Clustering methods use one distorted sample to compute 'targets' for the loss, and another distorted version of the sample to predict these targets, followed by an alternate optimization scheme like k-means in DEEPCLUSTER (Caron et al., 2018) or non-differentiable operators in SWAV (Caron et al., 2020) and SELA (Asano et al., 2020). In another recent line of work, BYOL (Grill et al., 2020) and SIMSIAM (Chen & He, 2020), both the network architecture and parameter updates are modified to introduce asymmetry. The network architecture is modified to be asymmetric using a special 'predictor' network and the parameter updates are asymmetric such that the model parameters are only updated using one distorted version of the input, while the representations from another distorted version are used as a fixed target. (Chen & He, 2020) conclude that the asymmetry of the learning update, 'stop-gradient', is critical to preventing trivial solutions.

In this paper, we propose a new method, BARLOW TWINS, which applies *redundancy-reduction* — a principle first proposed in neuroscience — to self-supervised learning. In his influential article *Possible Principles Underlying the Transformation of Sensory Messages* (Barlow, 1961), neuroscientist H. Barlow hypothesized that the goal of sensory processing is to recode highly redundant sensory inputs into a factorial code (a code with statistically independent components). This principle has been fruitful in explaining the organization of the visual system, from the retina to cortical areas (see (Barlow, 2001) for a review and (Lindsey et al., 2020; Ocko et al., 2018; Schwartz & Simoncelli, 2001) for recent efforts), and has led to a number of algorithms for supervised and unsupervised learning (Ballé et al., 2017; Deco & Parra, 1997; Földiák, 1990; Linsker, 1988; Redlich, 1993a;b; Schmidhuber et al., 1996). Based on this principle, we propose an objective function which tries to make the cross-correlation matrix computed from twin embeddings as close to the identity matrix as possible. BARLOW TWINS is conceptually simple, easy to implement and learns useful representations as opposed to trivial solutions. Compared to other methods, it does not require large batches (Chen et al., 2020a), nor does it require any asymmetric mechanisms like prediction networks (Grill et al., 2020), momentum encoders (He et al., 2019), non-differentiable operators (Caron et al., 2020) or stop-gradients (Chen & He, 2020). Intriguingly, BARLOW TWINS strongly benefits from the use of very high-dimensional embeddings. BARLOW TWINS outperforms previous methods on ImageNet for semi-supervised classification in the low-data regime (55% top-1 accuracy for 1% labels), and is on par with current state of the art for ImageNet classification with a linear classifier head, as well as for a number of transfer tasks of classification and object detection.

2. Method

2.1. Description of BARLOW TWINS

Like other methods for SSL (Caron et al., 2020; Chen et al., 2020a; Grill et al., 2020; He et al., 2019; Misra & van der Maaten, 2019), BARLOW TWINS operates on a joint embedding of distorted images (Fig. 1). More specifically, it produces two distorted views for all images of a batch X sampled from a dataset. The distorted views are obtained via a distribution of data augmentations $\mathcal{T}$. The two batches of distorted views Y^A and Y^B are then fed to a function f_θ, typically a deep network with trainable parameters θ, producing batches of embeddings Z^A and Z^B respectively. To simplify notations, Z^A and Z^B are assumed to be mean-centered along the batch dimension, such that each unit has mean output 0 over the batch.

BARLOW TWINS distinguishes itself from other methods by its innovative loss function $\mathcal{L}_{\mathcal{BT}}$:

$$\mathcal{L}_{\mathcal{BT}} \triangleq \underbrace{\sum_i (1-\mathcal{C}_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} \mathcal{C}_{ij}{}^2}_{\text{redundancy reduction term}} \tag{1}$$

where λ is a positive constant trading off the importance of the first and second terms of the loss, and where $\mathcal{C}$ is the cross-correlation matrix computed between the outputs of the two identical networks along the batch dimension:

$$\mathcal{C}_{ij} \triangleq \frac{\sum_b z^A_{b,i} z^B_{b,j}}{\sqrt{\sum_b \left(z^A_{b,i}\right)^2} \sqrt{\sum_b \left(z^B_{b,j}\right)^2}} \tag{2}$$

where b indexes batch samples and i, j index the vector dimension of the networks' outputs. $\mathcal{C}$ is a square matrix with size the dimensionality of the network's output, and with values comprised between -1 (i.e. perfect anti-correlation) and 1 (i.e. perfect correlation).

Intuitively, the *invariance term* of the objective, by trying to equate the diagonal elements of the cross-correlation matrix to 1, makes the embedding invariant to the distortions applied. The *redundancy reduction term*, by trying to equate the off-diagonal elements of the cross-correlation

通常通过孪生网络的变体（Hadsell et al., 2006），最大化同一样本不同扰动版本的表征相似性来实现。由于存在常量表征等平凡解，这些方法依靠不同机制学到有用表征。

SIMCLR 等对比方法（Chen et al., 2020a）定义“正”“负”样本对，并在损失中区别对待。它们还可以采用非对称学习更新，例如独立于主网络更新的动量编码器（He et al., 2019）。聚类方法用一个扰动样本计算损失的“目标”，再用同一样本的另一扰动版本预测目标，然后执行交替优化，例如 DEEPCLUSTER 的 k-means（Caron et al., 2018），或 SWAV（Caron et al., 2020）与 SELA（Asano et al., 2020）的不可微算子。另一类近期工作 BYOL（Grill et al., 2020）与 SIMSIAM（Chen & He, 2020）同时修改网络结构和参数更新，以引入非对称性。特殊“预测器”使结构非对称；更新的非对称性则表现为，模型参数仅通过输入的一个扰动版本更新，另一扰动版本的表征作为固定目标。Chen & He（2020）认为，这种学习更新的非对称性，即“停止梯度”，对避免平凡解至关重要。

本文提出 BARLOW TWINS，将最初源自神经科学的冗余消减原则用于自监督学习。在影响深远的《感觉信息转换背后可能的原则》（Barlow, 1961）一文中，神经科学家 H. Barlow 假设，感觉处理的目标是将高度冗余的感觉输入重新编码为因子化编码，即各分量统计独立的编码。该原则成功解释了从视网膜到皮层区域的视觉系统组织方式（综述见 Barlow, 2001；近期工作见 Lindsey et al., 2020; Ocko et al., 2018; Schwartz & Simoncelli, 2001），并催生了多种监督与无监督学习算法（Ball´e et al., 2017; Deco & Parra, 1997; F¨oldi´ak, 1990; Linsker, 1988; Redlich, 1993a;b; Schmidhuber et al., 1996）。基于这一原则，我们提出目标函数，使双网络嵌入的交叉相关矩阵尽可能接近单位矩阵。BARLOW TWINS 概念简单、易于实现，学到的是有用表征而非平凡解。与其他方法相比，它无需大批量（Chen et al., 2020a），也无需预测网络（Grill et al., 2020）、动量编码器（He et al., 2019）、不可微算子（Caron et al.,

2020）或停止梯度（Chen & He, 2020）等机制。有趣的是，BARLOW TWINS 明显受益于高维嵌入。在少标注的 ImageNet 半监督分类中，它优于以往方法（1%标注下 top-1 准确率55%）；在线性分类头的 ImageNet 分类，以及多种分类和检测迁移任务上，与当前最优方法相当。

2. 方法

2.1. BARLOW TWINS 方法描述

与其他自监督方法一样（Caron et al., 2020; Chen et al., 2020a; Grill et al., 2020; He et al., 2019; Misra & van der Maaten, 2019），BARLOW TWINS 作用于扰动图像的联合嵌入（图1）。具体而言，对于从数据集中抽取的一批图像 X，它为每张图像生成两个扰动视图。扰动由数据增强分布 $\mathcal{T}$ 产生。两批扰动视图 Y^A 和 Y^B 随后分别输入函数 f_θ；该函数通常是具有可训练参数 θ 的深度网络，分别输出两批嵌入 Z^A 和 Z^B。为简化记号，假设 Z^A 和 Z^B 已沿批量维度进行均值中心化，使每个单元在整个批次中的平均输出为0。

BARLOW TWINS 以创新的损失函数 $\mathcal{L}_{\mathcal{BT}}$ 区别于其他方法：

$$\mathcal{L}_{\mathcal{BT}} \triangleq \underbrace{\sum_i (1-\mathcal{C}_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} \mathcal{C}_{ij}{}^2}_{\text{redundancy reduction term}} \tag{1}$$

其中 λ 是权衡损失第一项与第二项重要性的正常数；$\mathcal{C}$ 是沿批量维度计算的两个相同网络输出之间的交叉相关矩阵：

$$\mathcal{C}_{ij} \triangleq \frac{\sum_b z^A_{b,i} z^B_{b,j}}{\sqrt{\sum_b \left(z^A_{b,i}\right)^2} \sqrt{\sum_b \left(z^B_{b,j}\right)^2}} \tag{2}$$

其中 b 索引批次样本，i、j 索引网络输出向量的维度。$\mathcal{C}$ 是边长等于网络输出维度的方阵，其值在−1（完全负相关）与1（完全正相关）之间。

直观地，不变性项使交叉相关矩阵的对角元素趋于1，从而让嵌入对输入扰动保持不变。冗余消减项使交叉相关

matrix to 0, decorrelates the different vector components of the embedding. This decorrelation reduces the redundancy between output units, so that the output units contain non-redundant information about the sample.

More formally, BARLOW TWINS's objective function can be understood through the lens of information theory, and specifically as an instanciation of the *Information Bottleneck (IB)* objective (Tishby & Zaslavsky, 2015; Tishby et al., 2000). Applied to self-supervised learning, the IB objective consists in finding a representation that conserves as much information about the sample as possible while being the *least* possible informative about the specific distortions applied to that sample. The mathematical connection between BARLOW TWINS's objective function and the IB principle is explored in Appendix A.

BARLOW TWINS' objective function has similarities with existing objective functions for SSL. For example, the redundancy reduction term plays a role similar to the *contrastive term* in the INFONCE objective (Oord et al., 2018), as discussed in detail in Section 5. However, important conceptual differences in these objective functions result in practical advantages of our method compared to INFONCE-based methods, namely that (1) our method does not require a large number of negative samples and can thus operate on small batches (2) our method benefits from very high-dimensional embeddings. Alternatively, the redundancy reduction term can be viewed as a *soft-whitening* constraint on the embeddings, connecting our method to a recently proposed method performing a *hard-whitening* operation on the embeddings (Ermolov et al., 2020), as discussed in Section 5. However, our method performs better than current hard-whitening methods.

The pseudocode for BARLOW TWINS is shown as Algorithm 1.

Algorithm 1 PyTorch-style pseudocode for Barlow Twins.

```
# f: encoder network
# lambda: weight on the off-diagonal terms
# N: batch size
# D: dimensionality of the embeddings
#
# mm: matrix-matrix multiplication
# off_diagonal: off-diagonal elements of a matrix
# eye: identity matrix

for x in loader: # load a batch with N samples
    # two randomly augmented versions of x
    y_a, y_b = augment(x)

    # compute embeddings
    z_a = f(y_a) # NxD
    z_b = f(y_b) # NxD

    # normalize repr. along the batch dimension
    z_a_norm = (z_a - z_a.mean(0)) / z_a.std(0) # NxD
    z_b_norm = (z_b - z_b.mean(0)) / z_b.std(0) # NxD

    # cross-correlation matrix
    c = mm(z_a_norm.T, z_b_norm) / N # DxD

    # loss
    c_diff = (c - eye(D)).pow(2) # DxD
    # multiply off-diagonal elems of c_diff by lambda
    off_diagonal(c_diff).mul_(lambda)
    loss = c_diff.sum()

    # optimization step
    loss.backward()
    optimizer.step()
```

2.2. Implementation Details

Image augmentations Each input image is transformed twice to produce the two distorted views shown in Figure 1. The image augmentation pipeline consists of the following transformations: random cropping, resizing to 224×224, horizontal flipping, color jittering, converting to grayscale, Gaussian blurring, and solarization. The first two transformations (cropping and resizing) are always applied, while the last five are applied randomly, with some probability. This probability is different for the two distorted views in the last two transformations (blurring and solarization). We use the same augmentation parameters as BYOL (Grill et al., 2020).

Architecture The encoder consists of a ResNet-50 network (He et al., 2016) (without the final classification layer, 2048 output units) followed by a projector network. The projector network has three linear layers, each with 8192 output units. The first two layers of the projector are followed by a batch normalization layer and rectified linear units. We call the output of the encoder the 'representations' and the output of the projector the 'embeddings'. The representations are used for downstream tasks and the embeddings are fed to the loss function of BARLOW TWINS.

Optimization We follow the optimization protocol described in BYOL (Grill et al., 2020). We use the LARS optimizer (You et al., 2017) and train for 1000 epochs with a batch size of 2048. We however emphasize that our model works well with batches as small as 256 (see Ablations). We use a learning rate of 0.2 for the weights and 0.0048 for the biases and batch normalization parameters. We multiply the learning rate by the batch size and divide it by 256. We use a learning rate warm-up period of 10 epochs, after which we reduce the learning rate by a factor of 1000 using a cosine decay schedule (Loshchilov & Hutter, 2016). We ran a search for the trade-off parameter λ of the loss function and found the best results for $\lambda = 5 \cdot 10^{-3}$. We use a weight decay parameter of $1.5 \cdot 10^{-6}$. The biases and batch normalization parameters are excluded from LARS adaptation and weight decay. Training is distributed across 32 V100 GPUs and takes approximately 124 hours. For comparison, our reimplementation of BYOL trained with a batch size of 4096 takes 113 hours on the same hardware.

矩阵的非对角元素趋于0，从而让嵌入的不同分量去相关。这减少了输出单元间的冗余，使各单元包含关于样本的非冗余信息。

更形式化地说，BARLOW TWINS 的目标可从信息论角度理解，尤其可视为信息瓶颈（IB）目标的一种实例（Tishby & Zaslavsky, 2015; Tishby et al., 2000）。在自监督学习中，IB 目标是找到一种表征，尽可能多地保留样本信息，同时尽可能少地携带该样本所施加的具体扰动信息。附录A探讨了 BARLOW TWINS 目标函数与 IB 原则的数学联系。

BARLOW TWINS 的目标函数与现有自监督目标函数有相似之处。例如，冗余消减项的作用类似于 InfoNCE 目标中的对比项（Oord et al., 2018），详见第5节。然而，这些目标在概念上的关键差异带来了相对于 InfoNCE 方法的实际优势：（1）本方法不需要大量负样本，因而可以使用小批量；（2）本方法受益于高维嵌入。也可将冗余消减项理解为对嵌入的软白化约束，这与近期直接对嵌入执行硬白化的方法有关（Ermolov et al., 2020），详见第5节。不过，本方法优于目前的硬白化方法。

BARLOW TWINS 的伪代码见算法1。

算法1 Barlow Twins 的 PyTorch 风格伪代码。

```
# f: encoder network
# lambda: weight on the off-diagonal terms
# N: batch size
# D: dimensionality of the embeddings
#
# mm: matrix-matrix multiplication
# off_diagonal: off-diagonal elements of a matrix
# eye: identity matrix

for x in loader: # load a batch with N samples
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    c = mm(z_a_norm.T, z_b_norm) / N # DxD

    # loss
    c_diff = (c - eye(D)).pow(2) # DxD
    # multiply off-diagonal elems of c_diff by lambda
    off_diagonal(c_diff).mul_(lambda)
    loss = c_diff.sum()

    # optimization step
    loss.backward()
    optimizer.step()
```

2.2. 实现细节

图像增强 每张输入图像都变换两次，产生图1中的两个扰动视图。增强流程包含随机裁剪、缩放到 224×224、水平翻转、颜色抖动、转为灰度、高斯模糊和曝光反转。前两项（裁剪与缩放）总会执行，后五项以一定概率随机执行。在最后两项（模糊与曝光反转）中，两个扰动视图采用不同的概率。我们采用与 BYOL 相同的增强参数（Grill et al., 2020）。

结构 编码器采用 ResNet-50（He et al., 2016），去除最终分类层，2048个输出单元），随后接投影网络。投影器有三层线性层，每层8192个输出单元，前两层后接批归一化和 ReLU。编码器的输出称为“表征”，投影器的输出称为“嵌入”。表征用于下游任务，嵌入输入 BARLOW TWINS 损失函数。

优化 我们遵循 BYOL 的优化流程（Grill et al., 2020），使用 LARS 优化器（You et al., 2017），以2048的批量大小训练1000轮。需强调，本模型在批量小至256时仍表现良好（见消融实验）。权重学习率为0.2，偏置与批归一化参数学习率为0.0048；学习率乘以批量大小再除以256。先进行10轮学习率预热，再用余弦衰减将学习率降低1000倍（Loshchilov & Hutter, 2016）。我们搜索了损失函数的权衡参数 λ，发现 $\lambda = 5 \cdot 10^{-3}$ 时结果最好。权重衰减参数为 $1.5 \cdot 10^{-6}$。偏置和批归一化参数不参与 LARS 自适应调整及权重衰减。训练分布在32块 V100 GPU 上，约需124小时。作为比较，我们复现的 BYOL 以4096的批量大小在相同硬件上训练需要113小时。

3. Results

We follow standard practice (Goyal et al., 2019) and evaluate our representations by transfer learning to different datasets and tasks in computer vision. Our network is pretrained using self-supervised learning on the training set of the ImageNet ILSVRC-2012 dataset (Deng et al., 2009) (without labels). We evaluate our model on a variety of tasks such as image classification and object detection, and using fixed representations from the network or finetuning it. We provide the hyperparameters for all the transfer learning experiments in the Appendix.

3.1. Linear and Semi-Supervised Evaluations on ImageNet

Linear evaluation on ImageNet We train a linear classifier on ImageNet on top of fixed representations of a ResNet-50 pretrained with our method. The top-1 and top-5 accuracies obtained on the ImageNet validation set are reported in Table 1. Our method obtains a top-1 accuracy of 73.2% which is comparable to the state-of-the-art methods.

Table 1. **Top-1 and top-5 accuracies (in %) under linear evaluation on ImageNet**. All models use a ResNet-50 encoder. Top-3 best self-supervised methods are underlined.

Method	Top-1	Top-5
Supervised	76.5	
MoCo	60.6	
PIRL	63.6	-
SimCLR	69.3	89.0
MoCo v2	71.1	90.1
SimSiam	71.3	-
SwAV (w/o multi-crop)	71.8	-
BYOL	74.3	91.6
SwAV	75.3	-
Barlow Twins (ours)	73.2	91.0

Semi-supervised training on ImageNet We fine-tune a ResNet-50 pretrained with our method on a subset of ImageNet. We use subsets of size 1% and 10% using the same split as SimCLR. The semi-supervised results obtained on the ImageNet validation set are reported in Table 2. Our method is either on par (when using 10% of the data) or slightly better (when using 1% of the data) than competing methods.

3.2. Transfer to other datasets and tasks

Image classification with fixed features We follow the setup from (Misra & van der Maaten, 2019) and train a linear classifier on fixed image representations, *i.e.*, the parameters of the ConvNet remain unchanged. We use a diverse set of datasets for this evaluation - Places-205 (Zhou et al., 2014) for scene classification, VOC07 (Everingham et al., 2010) for multi-label image classification, and iNaturalist2018 (Van Horn et al., 2018) for fine-grained image classification. We report our results in Table 3. Barlow Twins performs competitively against prior work, and outperforms SimCLR and MoCo-v2 on most datasets.

Table 2. **Semi-supervised learning on ImageNet** using 1% and 10% training examples. Results for the supervised method are from (Zhai et al., 2019). Best results are in **bold**.

Method	Top-1		Top-5	
	1%	10%	1%	10%
Supervised	25.4	56.4	48.4	80.4
PIRL	-	-	57.2	83.8
SimCLR	48.3	65.6	75.5	87.8
BYOL	53.2	68.8	78.4	89.0
SwAV	53.9	**70.2**	78.5	**89.9**
Barlow Twins (ours)	**55.0**	69.7	**79.2**	89.3

Table 3. **Transfer learning: image classification.** We benchmark learned representations on the image classification task by training linear classifiers on fixed features. We report top-1 accuracy on Places-205 and iNat18 datasets, and classification mAP on VOC07. Top-3 best self-supervised methods are underlined.

Method	Places-205	VOC07	iNat18
Supervised	53.2	87.5	46.7
SimCLR	52.5	85.5	37.2
MoCo-v2	51.8	86.4	38.6
SwAV (w/o multi-crop)	52.8	86.4	39.5
SwAV	56.7	88.9	48.6
BYOL	54.0	86.6	47.6
Barlow Twins (ours)	54.1	86.2	46.5

Object Detection and Instance Segmentation We evaluate our representations for the localization based tasks of object detection and instance segmentation. We use the VOC07+12 (Everingham et al., 2010) and COCO (Lin et al., 2014) datasets following the setup in (He et al., 2019) which finetunes the ConvNet parameters. Our results in Table 4 indicate that Barlow Twins performs comparably or better than state-of-the-art representation learning methods for these localization tasks.

3. 结果

我们遵循标准做法（Goyal et al., 2019），通过迁移到不同视觉数据集和任务来评估表征。网络在 ImageNet ILSVRC-2012 训练集（Deng et al., 2009）上进行无标注自监督预训练。评估涵盖图像分类和目标检测等任务，既使用固定表征，也微调网络。所有迁移实验的超参数见附录。

3.1. ImageNet 线性与半监督评估

ImageNet 线性评估 我们在本方法预训练的 ResNet-50 固定表征之上训练 ImageNet 线性分类器。验证集 top-1 与 top-5 准确率见表1。本方法 top-1 达 73.2%，与当前最优方法相当。

表1. ImageNet 线性评估的 top-1 和 top-5 准确率（%）。全部模型使用 ResNet-50 编码器，前三名自监督方法加下划线。

方法	Top-1	Top-5
监督学习	76.5	
MoCo	60.6	
PIRL	63.6	-
SIMCLR	69.3	89.0
MOCO V2	71.1	90.1
SIMSIAM	71.3	-
SWAV（无多裁剪）	71.8	-
BYOL	74.3	91.6
SWAV	75.3	-
BARLOW TWINS（本文）	73.2	91.0

ImageNet 半监督训练 我们在 ImageNet 子集上微调本方法预训练的 ResNet-50。采用1%及10%的子集，与 SIMCLR 使用相同划分。验证集上的半监督结果见表2：10%数据时与竞争方法相当，1%数据时略优。

3.2. 向其他数据集与任务迁移

固定特征的图像分类 遵循 Misra & van der Maaten（2019），在固定图像表征上训练线性分类器，即

表2. 用1%及10%训练样本进行 ImageNet 半监督学习。监督方法结果来自 Zhai et al.（2019），最佳结果加粗。

方法	Top-1		Top-5	
	1%	10%	1%	10%
监督学习	25.4	56.4	48.4	80.4
PIRL	-	-	57.2	83.8
SIMCLR	48.3	65.6	75.5	87.8
BYOL	53.2	68.8	78.4	89.0
SWAV	53.9	**70.2**	78.5	**89.9**
BARLOW TWINS（本文）	**55.0**	69.7	**79.2**	89.3

表3. 迁移学习：图像分类。在固定特征上训练线性分类器，评估所学表征。在 Places-205 和 iNat18 上报告 top-1 准确率，在 VOC07 上报告分类 mAP。前三名自监督方法加下划线。

方法	Places-205	VOC07	iNat18
监督学习	53.2	87.5	46.7
SimCLR	52.5	85.5	37.2
MoCo-v2	51.8	86.4	38.6
SWAV（无多裁剪）	52.8	86.4	39.5
SWAV	56.7	88.9	48.6
BYOL	54.0	86.6	47.6
BARLOW TWINS（本文）	54.1	86.2	46.5

卷积网络的参数保持不变。我们使用多样化数据集：Places-205（Zhou et al., 2014）用于场景分类，VOC07（Everingham et al., 2010）用于多标签图像分类，iNaturalist2018（Van Horn et al., 2018）用于细粒度分类。结果见表3。BARLOW TWINS 可媲美以往方法，在多数数据集上优于 SimCLR 和 MoCo-v2。

目标检测与实例分割 我们在这两类依赖定位的任务上评估表征。使用 VOC07+12（Everingham et al., 2010）及 COCO（Lin et al., 2014），遵循 He et al.（2019）的设置，微调卷积网络参数。表4表明，BARLOW TWINS 在这些定位任务上与最优表征学习方法相当或更好。

Table 4. **Transfer learning: object detection and instance segmentation.** We benchmark learned representations on the object detection task on VOC07+12 using Faster R-CNN (Ren et al., 2015) and on the detection and instance segmentation task on COCO using Mask R-CNN (He et al., 2017). All methods use the C4 backbone variant (Wu et al., 2019) and models on COCO are finetuned using the 1× schedule. Best results are in **bold**.

Method	VOC07+12 det			COCO det			COCO instance seg		
	AP_{all}	AP_{50}	AP_{75}	AP^{bb}	AP^{bb}_{50}	AP^{bb}_{75}	AP^{mk}	AP^{mk}_{50}	AP^{mk}_{75}
Sup.	53.5	81.3	58.8	38.2	58.2	41.2	33.3	54.7	35.2
MoCo-v2	**57.4**	82.5	**64.0**	**39.3**	58.9	**42.5**	**34.4**	55.8	36.5
SwAV	56.1	**82.6**	62.7	38.4	58.6	41.3	33.8	55.2	35.9
SimSiam	57	82.4	63.7	39.2	**59.3**	42.1	**34.4**	**56.0**	**36.7**
BT (ours)	56.8	**82.6**	63.4	39.2	59.0	**42.5**	34.3	**56.0**	36.5

4. Ablations

For all ablation studies, BARLOW TWINS was trained for 300 epochs instead of 1000 epochs in the previous section. A linear evaluation on ImageNet of this baseline model yielded a 71.4% top-1 accuracy and a 90.2% top-5 accuracy. For all the ablations presented we report the top-1 and top-5 accuracy of training linear classifiers on the 2048 dimensional `res5` features using the ImageNet train set.

Loss Function Ablations We alter our loss function (eqn. 1) in several ways to test the necessity of each term of the loss function, and to experiment with different practices popular in other loss functions for SSL, such as INFONCE. Table 5 recapitulates the different loss functions tested along with their results on a linear evaluation benchmark of Imagenet. First we find that removing the invariance term (on-diagonal term) or the redundancy reduction term (off-diagonal term) of our loss function leads to worse/-collapsed solutions, as expected. We then study the effect of different normalization strategies. We first try to normalize the embeddings along the feature dimension so that they lie on the unit sphere, as it is common practice for losses measuring a cosine similarity (Chen et al., 2020a; Grill et al., 2020; Wang & Isola, 2020). Specifically, we first normalize the embeddings along the batch dimension (with mean subtraction), then normalize the embeddings along the feature dimension (without mean subtraction), and finally we measure the (unnormalized) covariance matrix instead of the (normalized) cross-correlation matrix in eqn. 2. The performance is slightly reduced. Second, we try to remove batch-normalization operations in the two hidden layers of the projector network MLP. The performance is barely affected. Third, in addition to removing the batch-normalization in the hidden layers, we replace the cross-correlation matrix in eqn. 2 by the cross-covariance matrix (which means the features are no longer normalized along the batch dimension). The performance is substantially reduced. We finally try a cross-entropy loss with temperature, for which the on-diagonal term and off-diagonal term is controlled by a temperature hyperparameter τ and coefficient λ: $\mathcal{L} = -\log \sum_i \exp(\mathcal{C}_{ii}/\tau) + \lambda \log \sum_i \sum_{j \neq i} \exp(\max(\mathcal{C}_{ij}, 0)/\tau)$. The performance is reduced.

Table 5. **Loss function explorations**. We ablate the invariance and redundancy terms in our proposed loss and observe that both terms are necessary for good performance. We also experiment with different normalization schemes and a cross-entropy loss and observe reduced performance.

Loss function	Top-1	Top-5
Baseline	71.4	90.2
Only invariance term (on-diag term)	57.3	80.5
Only red. red. term (off-diag term)	0.1	0.5
Normalization along feature dim.	69.8	88.8
No BN in MLP	71.2	89.7
No BN in MLP + no Normalization	53.4	76.7
Cross-entropy with temp.	63.3	85.7

Robustness to Batch Size The INFONCE loss that draws negative examples from the minibatch suffer performance drops when the batch size is reduced (e.g. SIMCLR (Chen et al., 2020a)). We thus sought to test the robustness of BARLOW TWINS to small batch sizes. In order to adapt our model to different batch sizes, we performed a grid search on LARS learning rates for each batch size. We find that, unlike SIMCLR, our model is robust to small batch sizes (Fig. 2), with a performance almost unaffected for a batch as small as 256. In comparison the accuracy for SimCLR drops about 4 p.p. for batch size 256. This robustness to small batch size, also found in non-contrastive methods such as BYOL, further demonstrates that our method is not only conceptually (see Discussion) but also empirically different than the INFONCE objective.

Effect of Removing Augmentations We find that our model is not robust to removing some types of data augmentations, like SIMCLR but unlike BYOL (Fig. 3). While this can be seen as a disadvantage of our method compared to BYOL, it can also be argued that the representations learned by our method are better controlled by the specific set of distortions used, as opposed to BYOL for which the invariances learned seem generic and intriguingly independent of the specific distortions used.

Projector Network Depth & Width For other SSL methods, such as BYOL and SIMCLR, the projector network drastically reduces the dimensionality of the ResNet output.

表4. 迁移学习：目标检测与实例分割。用 Faster R-CNN（Ren et al., 2015）在 VOC07+12 上评估目标检测表征，并用 Mask R-CNN（He et al., 2017）在 COCO 上评估检测与实例分割。全部方法使用 C4 主干变体（Wu et al., 2019）；COCO 模型采用 1× 调度微调。最佳结果加粗。

方法	VOC07+12 检测			COCO 检测			COCO 实例分割		
	AP_{all}	AP_{50}	AP_{75}	AP^{bb}	AP^{bb}_{50}	AP^{bb}_{75}	AP^{mk}	AP^{mk}_{50}	AP^{mk}_{75}
Sup.	53.5	81.3	58.8	38.2	58.2	41.2	33.3	54.7	35.2
MoCo-v2	**57.4**	82.5	**64.0**	**39.3**	58.9	**42.5**	**34.4**	55.8	36.5
SwAV	56.1	**82.6**	62.7	38.4	58.6	41.3	33.8	55.2	35.9
SimSiam	57	82.4	63.7	39.2	**59.3**	42.1	**34.4**	**56.0**	**36.7**
BT（本文）	56.8	**82.6**	63.4	39.2	59.0	**42.5**	34.3	**56.0**	36.5

4. 消融实验

全部消融实验中，BARLOW TWINS 训练300轮，而非上一节的1000轮。基线模型在 ImageNet 线性评估中达到 top-1 71.4%、top-5 90.2%。所有消融均在2048维 res5 特征上，用 ImageNet 训练集训练线性分类器，报告 top-1 和 top-5 准确率。

损失函数消融 我们以多种方式修改损失函数（式1），检验各项是否必要，并尝试 InfoNCE 等自监督损失中的常用做法。表5汇总了不同损失及其 ImageNet 线性评估结果。首先，移除不变性项（对角项）或冗余消减项（非对角项），都会如预期导致更差的解或坍塌。随后研究归一化策略。我们首先沿特征维度归一化嵌入，使其位于单位球面上；这是余弦相似度损失的常用做法（Chen et al., 2020a; Grill et al., 2020; Wang & Isola, 2020）。具体而言，先沿批量维度归一化并减去均值，再沿特征维度归一化但不减去均值，最后用未归一化的协方差矩阵替代式2中归一化的交叉相关矩阵，性能略降。其次，移除投影器 MLP 两个隐藏层中的批归一化，性能几乎不受影响。第三，在移除隐藏层批归一化的同时，将式2的交叉相关矩阵换成交叉协方差矩阵，即不再沿批量维度归一化特征。性能显著降低。最后，我们尝试带温度的交叉熵损失，用温度超参数 τ 和系数 λ 控制对角项与非对角项：$\mathcal{L} = -\log \sum_i \exp(\mathcal{C}_{ii}/\tau) + \lambda \log \sum_i \sum_{j \neq i} \exp(\max(\mathcal{C}_{ij}, 0)/\tau)$。性能下降。

表5. 损失函数探索。我们分别移除所提损失中的不变性项和冗余项，发现两项对良好性能都必要。其他归一化方案及交叉熵损失均使性能下降。

损失函数	Top-1	Top-5
基线	71.4	90.2
仅不变性项（对角项）	57.3	80.5
仅冗余消减项（非对角项）	0.1	0.5
沿特征维度归一化	69.8	88.8
MLP 无 BN	71.2	89.7
MLP 无 BN + 无归一化	53.4	76.7
带温度的交叉熵	63.3	85.7

对批量大小的鲁棒性 从小批次中抽取负样本的 InfoNCE 损失，在减小批量时会出现性能下降，例如 SIMCLR （ Chen et al., 2020a ）。因此，我们检验 BARLOW TWINS 对小批量的鲁棒性。为适配不同批量，针对每种批量大小对 LARS 学习率进行网格搜索。我们发现，与 SIMCLR 不同，本模型对小批量具有鲁棒性（图2）；即使批量小至256，性能也几乎不变。相比之下，SimCLR 在批量256时准确率下降约4个百分点。BYOL 等非对比方法也表现出这种鲁棒性。这进一步表明，我们的方法与 InfoNCE 不仅在概念上不同（见讨论），在实验表现上也不同。

移除增强的影响 我们发现，与 SIMCLR 类似而与 BYOL 不同，本模型对移除某些数据增强并不鲁棒（图3）。相对于 BYOL，这可视为劣势；但也可以认为，本方法学到的表征更受所选扰动集合的控制，而 BYOL 所学不变性似乎较为通用，且令人费解地不依赖具体扰动。

投影器的深度与宽度 在 BYOL 和 SIMCLR 等其他自监督方法中，投影网络大幅降低 ResNet 输出的维度。

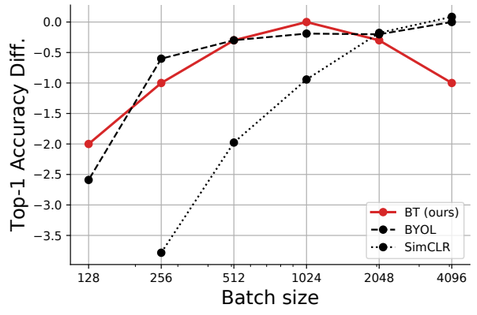


Figure 2. Effect of batch size. To compare the effect of the batch size across methods, for each method we report the difference between the top-1 accuracy at a given batch size and the best obtained accuracy among all batch size tested. BYOL: best accuracy is 72.5% for a batch size of 4096 (data from (Grill et al., 2020) fig. 3A). SIMCLR: best accuracy is 67.1% for a batch size of 4096 (data from (Chen et al., 2020a) fig. 9, model trained for 300 epochs). BARLOW TWINS: best accuracy is 71.7% for a batch size of 1024.

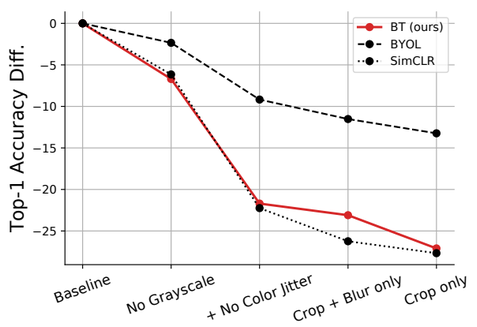


Figure 3. Effect of progressively removing data augmentations. Data for BYOL and SIMCLR (repro) is from (Grill et al., 2020) fig 3b.

In stark contrast, we find that BARLOW TWINS performs better when the dimensionality of the projector network output is very large. Other methods rapidly saturate when the dimensionality of the output increases, but our method keeps improving with all output dimensionality tested (Fig. 4). This result is quite surprising because the output of the ResNet is kept fixed to 2048, which acts as a dimensionality bottleneck in our model and sets the limit of the intrinsic dimensionality of the representation. In addition, similarly to other methods, we find that our model performs better when the projector network has more layers, with a saturation of the performance for 3 layers.

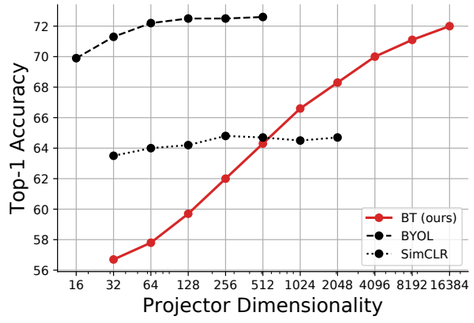


Figure 4. Effect of the dimensionality of the last layer of the projector network on performance. The parameter λ is kept fix for all dimensionalities tested. Data for SIMCLR is from (Chen et al., 2020a) fig 8; Data for BYOL is from (Grill et al., 2020) Table 14b.

Breaking Symmetry Many SSL methods (e.g. BYOL, SIMSIAM, SWAV) rely on different symmetry-breaking mechanisms to avoid trivial solutions. Our loss function avoids these trivial solutions by construction, even in the case of symmetric networks. It is however interesting to ask whether breaking symmetry can further improve the performance of our network. Following SIMSIAM and BYOL, we experiment with adding a predictor network composed of 2 fully connected layers of size 8192 to one of the network (with batch normalization followed by a ReLU nonlinearity in the hidden layer) and/or a stop-gradient mechanism on the other network. We find that these asymmetries slightly decrease the performance of our network (see Table 6).

Table 6. Effect of asymmetric settings

case	stop-gradient	predictor	Top-1	Top-5
Baseline	-	-	71.4	90.2
(a)	✓	-	70.5	89.0
(b)	-	✓	70.2	89.0
(c)	✓	✓	61.3	83.5

BYOL with a larger projector/predictor/embedding For a fair comparison with BYOL, we also evaluated BYOL with a wider and/or deeper projector head (3-layer MLP), a wider and/or deeper predictor head, and a larger dimensionality of the embedding. BYOL did not improve under these conditions (see Table 7).

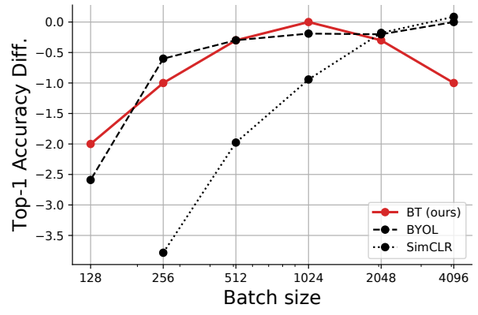


图2. 批量大小的影响。为跨方法比较，我们报告每种方法在给定批量大小下的 top-1 准确率，相对于其所有测试批量中的最佳准确率的差值。BYOL：批量4096时最佳准确率72.5%（数据来自 Grill et al., 2020，图3A）。SIMCLR：批量4096时最佳准确率67.1%（数据来自 Chen et al., 2020a，图9；模型训练300轮）。BARLOW TWINS：批量1024时最佳准确率71.7%。

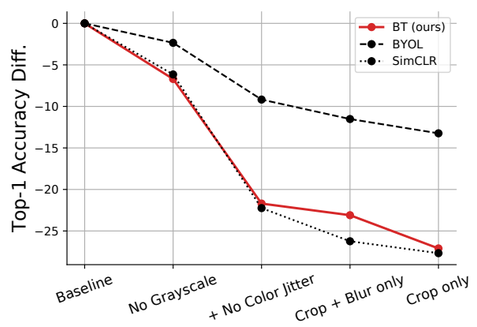


图3. 逐步移除数据增强的影响。BYOL 和 SIMCLR（复现）数据来自 Grill et al.（2020）图3b。

相比之下，BARLOW TWINS 在投影器输出维度很大时表现更好。其他方法随输出维度增大很快饱和，而本方法在全部测试维度下持续改善（图4）。这令人惊讶，因为 ResNet 输出始终固定为2048，构成模型的维度瓶颈，限制表征的内在维数。另外，与其他方法类似，我们发现模型表现更好，当投影网络层数更多时，性能在三层时饱和。

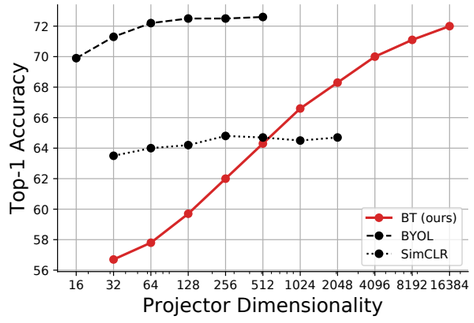


图4. 投影网络最后一层维度对性能的影响。全部测试维度采用固定参数 λ。SIMCLR 数据来自 Chen et al.（2020a）图8，BYOL 数据来自 Grill et al.（2020）表14b。

打破对称性 许多自监督方法（如 BYOL、SIMSIAM、SWAV）依靠不同的对称性破缺机制避免平凡解。我们的损失在构造上就能避免平凡解，即使网络对称也是如此。但打破对称性是否会进一步提高性能，仍值得探究。参考 SIMSIAM 与 BYOL，我们尝试为一个网络加入预测器：两层全连接层，维度8192，隐藏层中批归一化后接 ReLU；并且/或者为另一网络加入停止梯度。结果显示，这些非对称机制使性能略有下降（表6）。

表6. 非对称设置的影响

case	停止梯度	预测器	Top-1	Top-5
基线	-	-	71.4	90.2
(a)	✓	-	70.5	89.0
(b)	-	✓	70.2	89.0
(c)	✓	✓	61.3	83.5

使用更大投影器／预测器／嵌入的 BYOL 为公平比较，我们也评估了更宽和／或更深投影头（三层 MLP）、更宽和／或更深预测头，以及更高维嵌入的 BYOL。在这些设置下，BYOL 并未改善（表7）。

Table 7. Wider and/or deeper projector and predictor heads and larger dimensionality of the embedding did not improve the performance of BYOL.

Projector	Predictor	Acc1	Description
4096-256	4096-256	74.1%	baseline
4096-4096-256	4096-256	74.0%	3 layer proj, 2 layer pred, 256-d repr.
4096-4096-256	4096-4096-256	73.2%	3 layer proj, 3 layer pred, 256-d repr.
4096-4096-512	4096-512	73.7%	3 layer proj, 2 layer pred, 512-d repr.
4096-4096-512	4096-4096-512	73.2%	3 layer proj, 3 layer pred, 512-d repr.
8192-8192-8192	8192-8192	72.3%	same proj as BT, 2 layer pred, 8192-d repr.

Sensitivity to λ. We also explored the sensitivity of BARLOW TWINS to the hyperparameter λ, which trades off the desiderata of invariance and informativeness of the embeddings. We find that BARLOW TWINS is not very sensitive to this hyperparameter (Fig. 5).

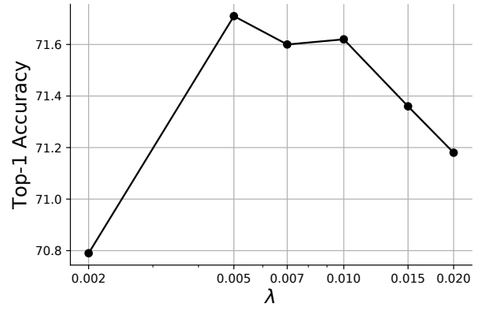


Figure 5. Sensitivity of BARLOW TWINS to the hyperparameter λ

5. Discussion

BARLOW TWINS learns self-supervised representations through a joint embedding of distorted images, with an objective function that maximizes similarity between the embedding vectors while reducing redundancy between their components. Our method does not require large batches of samples, nor does it require any particular asymmetry in the twin network structure. We discuss next the similarities and differences between our method and prior art, both from a conceptual and an empirical standpoint. For ease of comparison, all objective functions are recast with a common set of notations. The discussion ends with future directions.

5.1. Comparison with Prior Art

infoNCE The INFONCE loss, where NCE stands for Noise-Contrastive Estimation (Gutmann & Hyvärinen, 2010), is a popular type of contrastive loss function used for self-supervised learning (e.g. (Chen et al., 2020a; He et al., 2019; Hénaff et al., 2019; Oord et al., 2018)). It can be instantiated as:

$$\mathcal{L}_{infoNCE} \triangleq \underbrace{-\sum_b \frac{\langle z_b^A, z_b^B \rangle_i}{\tau \left\|z_b^A\right\|_2 \left\|z_b^B\right\|_2}}_{\text{similarity term}} + \underbrace{\sum_b \log \left(\sum_{b' \neq b} \exp \left(\frac{\langle z_b^A, z_{b'}^B \rangle_i}{\tau \left\|z_b^A\right\|_2 \left\|z_{b'}^B\right\|_2} \right) \right)}_{\text{contrastive term}}$$

where z^A and z^B are the twin network outputs, b indexes the sample in a batch, i indexes the vector component of the output, and τ is a positive constant called temperature in analogy to statistical physics.

For ready comparison, we rewrite BARLOW TWINS loss function with the same notations:

$$\mathcal{L}_{\mathcal{BT}} = \underbrace{\sum_i \left(1 - \frac{\langle z_{\cdot,i}^A, z_{\cdot,i}^B \rangle_b}{\left\|z_{\cdot,i}^A\right\|_2 \left\|z_{\cdot,i}^B\right\|_2} \right)^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} \left(\frac{\langle z_{\cdot,i}^A, z_{\cdot,j}^B \rangle_b}{\left\|z_{\cdot,i}^A\right\|_2 \left\|z_{\cdot,j}^B\right\|_2} \right)^2}_{\text{redundancy reduction term}}$$

Both BARLOW TWINS' and INFONCE's objective functions have two terms, the first aiming at making the embeddings invariant to the distortions fed to the twin networks, the second aiming at maximizing the variability of the embedding learned. Another common point between the two losses is that they both rely on batch statistics to measure this variability. However, the INFONCE objective maximizes the variability of the embeddings by maximizing the pairwise distance between all pairs of samples, whereas our method does so by decorrelating the components of the embeddings vectors.

The contrastive term in INFONCE can be interpreted as a non-parametric estimation of the entropy of the distribution of embeddings (Wang & Isola, 2020). An issue that arises with non-parametric entropy estimators is that they are prone to the curse of dimensionality: they can only be estimated reliably in a low-dimensional setting, and they typically require a large number of samples.

In contrast, our loss can be interpreted as a *proxy* entropy estimator of the distribution of embeddings under *a Gaussian parametrization* (see Appendix A). Thanks to this simplified parametrization, the variability of the embedding can be estimated from much fewer samples, and on very large-dimensional embeddings. Indeed, in the ablation studies that we perform, we find that (1) our method is robust to small batches unlike the popular INFONCE-based method

表7. 更宽和／或更深的投影头、预测头以及更高维嵌入，并未改善 BYOL 性能。

投影器	预测器	Acc1	描述
4096-256	4096-256	74.1%	基线
4096-4096-256	4096-256	74.0%	3层投影器，2层预测器，256维表征
4096-4096-256	4096-4096-256	73.2%	3层投影器，3层预测器，256维表征
4096-4096-512	4096-512	73.7%	3层投影器，2层预测器，512维表征
4096-4096-512	4096-4096-512	73.2%	3层投影器，3层预测器，512维表征
8192-8192-8192	8192-8192	72.3%	与 BT 相同的投影器，2层预测器，8192维表征

对 λ 的敏感性 我们还研究 BARLOW TWINS 对超参数 λ 的敏感性，它权衡嵌入的不变性与信息量。结果表明，本方法对该参数并不十分敏感（图5）。

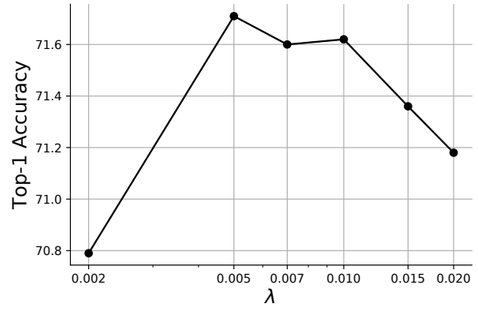


图5. BARLOW TWINS 对超参数 λ 的敏感性。

5. 讨论

BARLOW TWINS 通过扰动图像的联合嵌入学习自监督表征，目标函数最大化嵌入向量的相似性，同时减少各分量间的冗余。本方法无需大批量，也无需双网络结构中的特殊非对称性。下面从概念和实验两方面讨论它与以往方法的异同。为便于比较，所有目标函数都改写为统一记号。最后讨论未来方向。

5.1. 与以往方法的比较

InfoNCE InfoNCE 中的 NCE 指噪声对比估计（Gutmann & Hyv¨arinen, 2010），是一种常用于自监督学习的对比损失（例如 Chen et al., 2020a; He et al., 2019; H´enaff et al., 2019; Oord et al., 2018）。它可以具体写为：

$$\mathcal{L}_{infoNCE} \triangleq \underbrace{-\sum_b \frac{\langle z_b^A, z_b^B \rangle_i}{\tau \left\|z_b^A\right\|_2 \left\|z_b^B\right\|_2}}_{\text{similarity term}} + \underbrace{\sum_b \log \left(\sum_{b' \neq b} \exp \left(\frac{\langle z_b^A, z_{b'}^B \rangle_i}{\tau \left\|z_b^A\right\|_2 \left\|z_{b'}^B\right\|_2} \right) \right)}_{\text{contrastive term}}$$

其中 z^A 和 z^B 为双网络输出，b 索引批次样本，i 索引输出向量分量；τ 是正常数，类比统计物理称为温度。

为便于比较，用相同记号改写 BARLOW TWINS 损失：

$$\mathcal{L}_{\mathcal{BT}} = \underbrace{\sum_i \left(1 - \frac{\langle z_{\cdot,i}^A, z_{\cdot,i}^B \rangle_b}{\left\|z_{\cdot,i}^A\right\|_2 \left\|z_{\cdot,i}^B\right\|_2} \right)^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} \left(\frac{\langle z_{\cdot,i}^A, z_{\cdot,j}^B \rangle_b}{\left\|z_{\cdot,i}^A\right\|_2 \left\|z_{\cdot,j}^B\right\|_2} \right)^2}_{\text{redundancy reduction term}}$$

BARLOW TWINS 与 InfoNCE 的目标都有两项：第一项使嵌入对双网络输入的扰动保持不变，第二项最大化所学嵌入的变化程度。二者都依靠批次统计量衡量这种变化。然而，InfoNCE 通过最大化所有样本对的成对距离实现这一目标；我们则通过让嵌入向量各分量去相关来实现。

InfoNCE 的对比项可解释为嵌入分布熵的非参数估计（Wang & Isola, 2020）。非参数熵估计容易受到维数灾难影响：只能在低维情况下可靠估计，通常需要大量样本。

相比之下，我们的损失可解释为高斯参数化下嵌入分布熵的替代估计器（见附录A）。这一简化参数化允许用少得多的样本估计高维嵌入的变化程度。消融实验确实发现：（1）与常见的 InfoNCE 方法不同，本方法对小批量具有鲁棒性

SIMCLR, and (2) our method benefits from using very large dimensional embeddings, unlike INFONCE-based methods which do not see a benefit in increasing the dimensionality of the output.

Our loss presents several other interesting differences with infoNCE:

- In INFONCE, the embeddings are typically normalized along the feature dimension to compute a cosine similarity between embedded samples. We normalize the embeddings along the batch dimension instead.
- In our method, there is a parameter λ that trades off how much emphasis is put on the invariance term vs. the redundancy reduction term. This parameter can be interpreted as the trade-off parameter in the *Information Bottleneck* framework (see Appendix A). This parameter is not present in INFONCE.
- INFONCE also has a hyperparameter, the temperature, which can be interpreted as the width of the kernel in a non-parametric kernel density estimation of entropy, and practically weighs the relative importance of the hardest negative samples present in the batch (Chen et al., 2020a).

A number of alternative methods to ours have been proposed to alleviate the reliance on large batches of the INFONCE loss. For example, MoCo (Chen et al., 2020b; He et al., 2019) builds a dynamic dictionary of negative samples with a queue and a moving-averaged encoder. This enables building a large and consistent dictionary on-the-fly that facilitates contrastive unsupervised learning. MoCo typically needs to store $> 60{,}000$ sample embeddings. In contrast, our method does not require such a large dictionary, since it works well with a relatively small batch size (e.g. 256).

Asymmetric Twins BOOTSTRAP-YOUR-OWN-LATENT (aka BYOL) (Grill et al., 2020) and SIMSIAM (Chen & He, 2020) are two recent methods which use a simple cosine similarity between twin embeddings as an objective function, without *any* contrastive term:

$$\mathcal{L}_{cosine} = -\sum_b \frac{\langle z_b^A, z_b^B \rangle_i}{\|z_b^A\|_2 \, \|z_b^B\|_2}$$

Surprisingly, these methods successfully avoid trivial solutions by introducing some asymmetry in the architecture and learning procedure of the twin networks. For example, BYOL uses a predictor network which breaks the symmetry between the two networks, and also enforces an exponential moving average on the target network weights to slow down the progression of the weights on the target network. Combined together, these two mechanisms surprisingly avoid trivial solutions. The reasons behind this success are the subject of recent theoretical and empirical studies (Chen & He, 2020; Fetterman & Albrecht, 2020; Richemond et al., 2020; Tian et al., 2020). In particular, the ablation study (Chen & He, 2020) shows that the moving average is not necessary, but that stop-gradient on one of the branch and the presence of the predictor network are two crucial elements to avoid collapse. Other works show that batch normalization (Fetterman & Albrecht, 2020; Tian et al., 2020) or alternatively group normalization (Richemond et al., 2020) could play an important role in avoiding collapse.

Like our method, these asymmetric methods do not require large batches, since in their case there is no interaction between batch samples in the objective function.

It should be noted however that these asymmetric methods cannot be described as the optimization of an overall learning objective. Instead, there exists trivial solutions to the learning objective that these methods avoid via particular implementation choices and/or the result of non-trivial learning dynamics. In contrast, our method avoids trivial solutions by construction, making our method conceptually simpler and more principled than these alternatives (until their principle is discovered, see (Tian et al., 2021) for an early attempt).

Whitening In a concurrent work, (Ermolov et al., 2020) propose W-MSE. Acting on the embeddings from identical twin networks, this method performs a differentiable whitening operation (via Cholesky decomposition) of each batch of embeddings before computing a simple cosine similarity between the whitened embeddings of the twin networks. In contrast, the redundancy reduction term in our loss encourages the whitening of the batch embeddings as a soft constraint. The current W-MSE model achieves 66.3% top-1 accuracy on the Imagenet linear evaluation benchmark. It is an interesting direction for future studies to determine whether improved versions of this hard-whitening strategy could also lead to state-of-the-art results on these large-scale computer vision benchmarks.

Clustering These methods, such as DEEPCLUSTER (Caron et al., 2018), SWAV (Caron et al., 2020), SELA (Asano et al., 2020), perform contrastive-like comparisons without the requirement to compute all pairwise distances. Specifically, these methods simultaneously cluster the data while enforcing consistency between cluster assignments produced for different distortions of the same image, instead of comparing features directly as in contrastive learning. Clustering methods are also prone to collapse, *e.g.*, empty clusters in k-means and avoiding them relies on careful implementation details. Online clustering methods like SWAV

SIMCLR；（2）本方法受益于高维嵌入，而InfoNCE 方法并不因增加输出维度而改善。

我们的损失与 InfoNCE 还有其他有趣差异：

- InfoNCE 通常沿特征维度归一化嵌入，以计算样本嵌入间的余弦相似度。我们则沿批量维度归一化嵌入。
- 我们的方法有参数 λ，用来权衡不变性项与冗余消减项的重视程度。它可解释为信息瓶颈框架中的权衡参数（见附录A）。InfoNCE 没有这一参数。
- InfoNCE 也有超参数，即温度。它可解释为熵的非参数核密度估计中的核宽度；实际中，它调节批次里最难负样本的相对重要性（Chen et al., 2020a）。

已有多种替代方法缓解 InfoNCE 对大批量的依赖。例如 MoCo（Chen et al., 2020b; He et al., 2019）通过队列和移动平均编码器构建动态负样本字典，能在线建立大型且一致的字典，促进对比式无监督学习。MoCo 通常需要存储 $> 60{,}000$ 个样本嵌入。我们的方法无需如此大的字典，因为较小批量（如256）就已表现良好。

非对称双网络 BYOL（BOOTSTRAP-YOUR-OWN-LATENT；Grill et al., 2020）和 SIMSIAM（Chen & He, 2020）是两种近期方法，以双网络嵌入之间的简单余弦相似度作为目标，不含对比项：

$$\mathcal{L}_{cosine} = -\sum_b \frac{\langle z_b^A, z_b^B \rangle_i}{\|z_b^A\|_2 \, \|z_b^B\|_2}$$

令人意外的是，它们通过在双网络结构和学习过程中引入某种非对称性，成功避免平凡解。例如 BYOL 的预测器打破网络间对称性，并对目标网络权重施加指数移动平均，以减缓目标网络权重的变化过程。这两个机制结合后，出人意料地避免了平凡解。其成功原因是近期理论与实证研究的主题（Chen & He, 2020; Fetterman & Albrecht, 2020; Richemond et al., 2020; Tian et al., 2020）。尤其是 Chen & He（2020）的消融表明，移动平均并非必要，但一个分支的停止梯度和预测网络的存在，是避免坍塌的两个关键要素。其他工作表明，批归一化（Fetterman & Albrecht, 2020; Tian et al., 2020），或者组归一化（Richemond et al., 2020），可能对避免坍塌发挥重要作用。

与本方法一样，这些非对称方法无需大批量，因为其目标函数中没有批次样本之间的相互作用。

不过，应注意这些非对称方法不能被描述为优化一个整体学习目标。其目标存在平凡解，只是借助特定实现选择及/或复杂学习动态才避开它们。相比之下，本方法在构造上避免平凡解，因此概念更简单，原则更明确；至少在那些替代方法的原则尚未被发现之前如此（早期尝试见 Tian et al., 2021）。

白化 同期工作 Ermolov et al.（2020）提出 W-MSE。该方法处理相同双网络的嵌入：先通过 Cholesky 分解，对每批嵌入执行可微白化，再计算双网络白化嵌入之间的简单余弦相似度。相比之下，我们损失中的冗余消减项通过软约束鼓励批量嵌入白化。目前 W-MSE 在 ImageNet 线性评估中的 top-1 准确率为66.3%。研究硬白化策略的改进版本是否也能在这些大型视觉基准上达到最优水平，是值得探索的方向。

聚类 DEEPCLUSTER（Caron et al., 2018）、SWAV（Caron et al., 2020）、SELA（Asano et al., 2020）等方法进行类似对比的比较，却无需计算所有成对距离。具体而言，它们同时对数据聚类，并要求同一图像不同扰动版本的聚类分配一致；而对比学习直接比较特征。聚类方法也容易坍塌，例如 k-means 的空聚类；避免坍塌依赖精心设计的实现细节。SWAV 等在线聚类方法

can be trained with large and small batches but require storing features when the number of clusters is much larger than the batch size. Clustering methods can also be combined with contrastive learning (Li et al., 2021) to prevent collapse.

Noise As Targets This method (Bojanowski & Joulin, 2017) learns to map samples to fixed random targets on the unit sphere, which can be interpreted as a form of whitening. This objective uses a single network, and hence does not leverage the distortions induced by twin networks. Predefining random targets might limit the flexibility of the representation that can be learned.

IMAX In the early days of SSL, (Becker & Hinton, 1992; Zemel & Hinton, 1990) proposed a loss function between twin networks given by:

$$\mathcal{L}_{IMAX} \triangleq \log|\mathcal{C}_{(Z^A-Z^B)}| - \log|\mathcal{C}_{(Z^A+Z^B)}|$$

where $||$ denotes the determinant of a matrix, $\mathcal{C}_{(Z^A-Z^B)}$ is the covariance matrix of the difference of the outputs of the twin networks and $\mathcal{C}_{(Z^A+Z^B)}$ the covariance of the sum of these outputs. It can be shown that this objective maximizes the information between the twin network representations under the assumptions that the two representations are noisy versions of the same underlying Gaussian signal, and that the noise is independant, additive and Gaussian. This objective is similar to ours in the sense that there is one term that encourages the two representations to be similar and another term that encourages the units to be decorrelated. However, unlike IMAX, our objective is not directly an information quantity, and we have an extra trade-off parameter λ that trades off the two terms of our loss. The IMAX objective was used in early work so it is not clear whether it can scale to large computer vision tasks. Our attempts to make it work on ImageNet were not successful.

5.2. Future Directions

We observe a steady improvement of the performance of our method as we increase the dimensionality of the embeddings (i.e. of the last layer of the projector network). This intriguing result is in stark contrast with other popular methods for SSL, such as SIMCLR (Chen et al., 2020a) and BYOL (Grill et al., 2020), for which increasing the dimensionality of the embeddings rapidly saturates performance. It is a promising avenue to continue this exploration for even higher dimensional embeddings ($> 16{,}000$), but this would require the development of new methods or alternative hardware to accommodate the memory requirements of operating on such large embeddings.

Our method is just one possible instanciation of the *Information Bottleneck* principle applied to SSL. We believe that further refinements of the proposed loss function and algorithm could lead to more efficient solutions and even better performances. For example, the redundancy reduction term is currently computed from the off-diagonal terms of the cross-correlation matrix between the twin network embeddings, but alternatively it could be computed from the off-diagonal terms of the auto-correlation matrix of a single network's embedding. Our preliminary analyses seem to indicate that this alternative leads to similar performances (not shown). A modified loss could also be applied to the (unnormalized) cross-covariance matrix instead of the (normalized) cross-correlation matrix (see Ablations for preliminary analyses).

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References

Asano, Y. M., Rupprecht, C., and Vedaldi, A. Self-labelling via simultaneous clustering and representation learning. *International Conference on Learning Representations (ICLR)*, 2020.

Ballé, J., Laparra, V., and Simoncelli, E. P. End-to-end Optimized Image Compression. In *International Conference on Learning Representations (ICLR)*, 2017.

Barlow, H. Redundancy reduction revisited. *Network (Bristol, England)*, 12(3):241–253, August 2001. ISSN 0954-898X.

Barlow, H. B. *Possible Principles Underlying the Transformations of Sensory Messages*. in Sensory Communication, The MIT Press, 1961.

Becker, S. and Hinton, G. E. Self-organizing neural network that discovers surfaces in random-dot stereograms. *Nature*, 355(6356):161–163, January 1992.

Bojanowski, P. and Joulin, A. Unsupervised learning by predicting noise. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2017.

Cai, T. T., Liang, T., and Zhou, H. H. Law of Log Determinant of Sample Covariance Matrix and Optimal Estimation of Differential Entropy for High-Dimensional Gaussian Distributions. *Journal of Multivariate Analysis*, 137, 2015. doi: 10.1016/j.jmva.2015.02.003.

Caron, M., Bojanowski, P., Joulin, A., and Douze, M. Deep clustering for unsupervised learning of visual features. In

可以用大批量或小批量训练，但当聚类数量远大于批量大小时，需要存储特征。聚类方法也可与对比学习结合以防坍塌（Li et al., 2021）。

以噪声为目标 该方法（Bojanowski & Joulin, 2017）学习将样本映射到单位球面上的固定随机目标，可视为一种白化。它只使用一个网络，因此没有利用双网络产生的扰动。预定义随机目标可能限制所学表征的灵活性。

IMAX 在自监督学习早期，Becker & Hinton（1992）及 Zemel & Hinton（1990）提出双网络损失：

$$\mathcal{L}_{IMAX} \triangleq \log|\mathcal{C}_{(Z^A-Z^B)}| - \log|\mathcal{C}_{(Z^A+Z^B)}|$$

其中 $||$ 表示矩阵的行列式，$\mathcal{C}_{(Z^A-Z^B)}$ 是双网络输出之差的协方差矩阵，$\mathcal{C}_{(Z^A+Z^B)}$ 是输出之和的协方差矩阵。假设两种表征是同一底层高斯信号的含噪版本，噪声独立、可加且服从高斯分布，则可证明该目标最大化双网络表征之间的信息。它与我们的目标相似：一项鼓励两种表征相似，另一项鼓励各单元去相关。然而，与 IMAX 不同，我们的目标并不直接是一种信息量，而且额外引入参数 λ，权衡损失的两项。IMAX 用于早期工作，因此尚不清楚它能否扩展到大型计算机视觉任务；我们在 ImageNet 上的尝试未获成功。

5.2. 未来方向

随着嵌入维度（即投影器最后一层维度）增加，我们的方法性能持续提高。这与 SIMCLR（Chen et al., 2020a）和 BYOL（Grill et al., 2020）等方法形成鲜明差异：它们的嵌入维度增大后，性能很快饱和。继续探索更高维嵌入（$> 16{,}000$）很有前景，但如此大的嵌入需要更多内存，因此必须开发新方法或采用其他硬件。

本方法只是 IB 原则用于自监督学习的一种实例。我们相信，进一步改进所提损失函数和算法可能带来更高效的解及更佳性能。例如，目前冗余消减项由双网络嵌入的交叉相关矩阵非对角项计算；也可以改用单网络嵌入自相关矩阵的非对角项。初步分析表明，后一方案性能相近（未展示）。修改后的损失也可用于未归一化的交叉协方差矩阵，替代归一化的交叉相关矩阵（初步分析见消融实验）。

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参考文献

Asano, Y. M., Rupprecht, C., and Vedaldi, A. Self-labelling via simultaneous clustering and representation learning. International Conference on Learning Representations (ICLR), 2020. Ball´e, J., Laparra, V., and Simoncelli, E. P. End-to-end Optimized Image Compression. In International Conference on Learning Representations (ICLR), 2017. Barlow, H. Redundancy reduction revisited. Network (Bris- tol, England), 12(3):241–253, August 2001. ISSN 0954- 898X. Barlow, H. B. Possible Principles Underlying the Transfor- mations of Sensory Messages. in Sensory Communica- tion, The MIT Press, 1961. Becker, S. and Hinton, G. E. Self-organizing neural net- work that discovers surfaces in random-dot stereograms. Nature, 355(6356):161–163, January 1992. Bojanowski, P. and Joulin, A. Unsupervised learning by predicting noise. In Proceedings of the International Conference on Machine Learning (ICML), 2017. Cai, T. T., Liang, T., and Zhou, H. H. Law of Log Determinant of Sample Covariance Matrix and Optimal Es- timation of Differential Entropy for High-Dimensional Gaussian Distributions. Journal of Multivariate Analysis, 137, 2015. doi: 10.1016/j.jmva.2015.02.003.

Caron, M., Bojanowski, P., Joulin, A., and Douze, M. Deep clustering for unsupervised learning of visual features. In

Proceedings of the European Conference on Computer Vision (ECCV), 2018.

Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., and Joulin, A. Unsupervised learning of visual features by contrasting cluster assignments. In *NeurIPS*, 2020.

Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. A simple framework for contrastive learning of visual representations. *arXiv preprint arXiv:2002.05709*, 2020a.

Chen, X. and He, K. Exploring simple siamese representation learning. *arXiv preprint arXiv:2011.10566*, 2020.

Chen, X., Fan, H., Girshick, R., and He, K. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020b.

Deco, G. and Parra, L. Non-linear Feature Extraction by Redundancy Reduction in an Unsupervised Stochastic Neural Network. *Neural Networks*, 10(4):683–691, June 1997. ISSN 0893-6080. doi: 10.1016/S0893-6080(96)00110-4.

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. In *Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009.

Ermolov, A., Siarohin, A., Sangineto, E., and Sebe, N. Whitening for Self-Supervised Representation Learning. *arXiv preprint arXiv:2007.06346*, 2020.

Everingham, M., Van Gool, L., Williams, C. K., Winn, J., and Zisserman, A. The pascal visual object classes (voc) challenge. *International journal of computer vision*, 88 (2):303–338, 2010.

Fetterman, A. and Albrecht, J. Understanding self-supervised and contrastive learning with "Bootstrap Your Own Latent" (BYOL). *Untitled AI*, August 2020.

Földiák, P. Forming sparse representations by local anti-Hebbian learning. *Biological Cybernetics*, 64(2):165–170, December 1990. ISSN 1432-0770. doi: 10.1007/BF02331346.

Goyal, P., Dollár, P., Girshick, R., Noordhuis, P., Wesolowski, L., Kyrola, A., Tulloch, A., Jia, Y., and He, K. Accurate, large minibatch sgd: Training imagenet in 1 hour. *arXiv preprint arXiv:1706.02677*, 2017.

Goyal, P., Mahajan, D., Gupta, A., and Misra, I. Scaling and benchmarking self-supervised visual representation learning. *Proceedings of the International Conference on Computer Vision (ICCV)*, 2019.

Grill, J.-B., Strub, F., Altché, F., Tallec, C., Richemond, P. H., Buchatskaya, E., Doersch, C., Pires, B. A., Guo, Z. D., Azar, M. G., et al. Bootstrap your own latent: A new approach to self-supervised learning. In *NeurIPS*, 2020.

Gutmann, M. and Hyvärinen, A. Noise-contrastive estimation: A new estimation principle for unnormalized statistical models. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*. JMLR Workshop and Conference Proceedings, 2010.

Hadsell, R., Chopra, S., and LeCun, Y. Dimensionality reduction by learning an invariant mapping. In *Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR)*, 2006.

He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

He, K., Gkioxari, G., Dollár, P., and Girshick, R. Mask r-cnn. In *Proceedings of the International Conference on Computer Vision (ICCV)*, 2017.

He, K., Fan, H., Wu, Y., Xie, S., and Girshick, R. Momentum contrast for unsupervised visual representation learning. *arXiv preprint arXiv:1911.05722*, 2019.

Hénaff, O. J., Razavi, A., Doersch, C., Eslami, S., and Oord, A. v. d. Data-efficient image recognition with contrastive predictive coding. *arXiv preprint arXiv:1905.09272*, 2019.

Li, Y., Hu, P., Liu, Z., Peng, D., Zhou, J. T., and Peng, X. Contrastive clustering. In *AAAI*, 2021.

Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., and Zitnick, C. L. Microsoft coco: Common objects in context. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2014.

Lindsey, J., Ocko, S. A., Ganguli, S., and Deny, S. A unified theory of early visual representations from retina to cortex through anatomically constrained deep CNNs. In *ICLR*, 2020.

Linsker, R. Self-organization in a perceptual network. *Computer*, 21(3):105–117, March 1988. ISSN 1558-0814. doi: 10.1109/2.36. Conference Name: Computer.

Loshchilov, I. and Hutter, F. Sgdr: Stochastic gradient descent with warm restarts. *arXiv preprint arXiv:1608.03983*, 2016.

Misra, I. and van der Maaten, L. Self-supervised learning of pretext-invariant representations. *arXiv preprint arXiv:1912.01991*, 2019.

Proceedings of the European Conference on Computer Vision(ECCV), 2018.

Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., and Joulin, A. Unsupervised learning of visual features by contrasting cluster assignments. In NeurIPS, 2020.

Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. A simple framework for contrastive learning of visual representations. arXiv preprint arXiv:2002.05709, 2020a.

Chen, X. and He, K. Exploring simple siamese representation learning. arXiv preprint arXiv:2011.10566, 2020.

Chen, X., Fan, H., Girshick, R., and He, K. Improved baselines with momentum contrastive learning. arXiv preprint arXiv:2003.04297, 2020b. Deco, G. and Parra, L. Non-linear Feature Extraction by Redundancy Reduction in an Unsupervised Stochastic Neural Network. Neural Networks, 10(4):683–691, June 1997. ISSN 0893-6080. doi: 10.1016/S0893-6080(96) 00110-4. Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. In Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR), 2009. Ermolov, A., Siarohin, A., Sangineto, E., and Sebe, N. Whitening for Self-Supervised Representation Learning. arXiv preprint arXiv:2007.06346, 2020. Everingham, M., Van Gool, L., Williams, C. K., Winn, J., and Zisserman, A. The pascal visual object classes (voc) challenge. International journal of computer vision, 88 (2):303–338, 2010. Fetterman, A. and Albrecht, J. Understanding self- supervised and contrastive learning with "Bootstrap Your Own Latent" (BYOL). Untitled AI, August 2020. F¨oldi´ak, P. Forming sparse representations by local anti-Hebbian learning. Biological Cybernetics, 64(2):165–170, December 1990. ISSN 1432-0770. doi: 10.1007/BF02331346. Goyal, P., Doll´ar, P., Girshick, R., Noordhuis, P., Wesolowski, L., Kyrola, A., Tulloch, A., Jia, Y., and He, K. Accurate, large minibatch sgd: Training imagenet in 1 hour. arXiv preprint arXiv:1706.02677, 2017. Goyal, P., Mahajan, D., Gupta, A., and Misra, I. Scaling and benchmarking self-supervised visual representation learning. Proceedings of the International Conference on Computer Vision (ICCV), 2019.

Grill, J.-B., Strub, F., Altch´e, F., Tallec, C., Richemond, P. H., Buchatskaya, E., Doersch, C., Pires, B. A., Guo, Z. D., Azar, M. G., et al. Bootstrap your own latent: A new approach to self-supervised learning. In NeurIPS, 2020.

Gutmann, M. and Hyv¨arinen, A. Noise-contrastive estimation: A new estimation principle for unnormalized statis- tical models. In Proceedings of the Thirteenth Interna- tional Conference on Artificial Intelligence and Statistics. JMLR Workshop and Conference Proceedings, 2010.

Hadsell, R., Chopra, S., and LeCun, Y. Dimensionality reduction by learning an invariant mapping. In Proceed- ings of the Conference on Computer Vision and Pattern Recognition (CVPR), 2006. He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR), 2016. He, K., Gkioxari, G., Doll´ar, P., and Girshick, R. Mask r-cnn. In Proceedings of the International Conference on Computer Vision (ICCV), 2017. He, K., Fan, H., Wu, Y., Xie, S., and Girshick, R. Momentum contrast for unsupervised visual representation learning. arXiv preprint arXiv:1911.05722, 2019. H´enaff, O. J., Razavi, A., Doersch, C., Eslami, S., and Oord, A. v. d. Data-efficient image recognition with contrastive predictive coding. arXiv preprint arXiv:1905.09272, 2019. Li, Y., Hu, P., Liu, Z., Peng, D., Zhou, J. T., and Peng, X. Contrastive clustering. In AAAI, 2021. Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Doll´ar, P., and Zitnick, C. L. Microsoft coco: Common objects in context. In Proceedings of the European Conference on Computer Vision (ECCV), 2014. Lindsey, J., Ocko, S. A., Ganguli, S., and Deny, S. A unified theory of early visual representations from retina to cortex through anatomically constrained deep CNNs. In ICLR, 2020. Linsker, R. Self-organization in a perceptual network. Com- puter, 21(3):105–117, March 1988. ISSN 1558-0814. doi: 10.1109/2.36. Conference Name: Computer. Loshchilov, I. and Hutter, F. Sgdr: Stochastic gra- dient descent with warm restarts. arXiv preprint arXiv:1608.03983, 2016. Misra, I. and van der Maaten, L. Self-supervised learn- ing of pretext-invariant representations. arXiv preprint arXiv:1912.01991, 2019.

Ocko, S., Lindsey, J., Ganguli, S., and Deny, S. The emergence of multiple retinal cell types through efficient coding of natural movies. In *NeurIPS*, 2018.

Oord, A. v. d., Li, Y., and Vinyals, O. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

Redlich, A. N. Redundancy Reduction as a Strategy for Unsupervised Learning. *Neural Computation*, 5(2):289–304, March 1993a. ISSN 0899-7667. doi: 10.1162/neco.1993.5.2.289. Conference Name: Neural Computation.

Redlich, A. N. Supervised Factorial Learning. *Neural Computation*, 5(5):750–766, September 1993b. ISSN 0899-7667, 1530-888X. doi: 10.1162/neco.1993.5.5.750.

Ren, S., He, K., Girshick, R., and Sun, J. Faster r-cnn: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

Richemond, P. H., Grill, J.-B., Altché, F., Tallec, C., Strub, F., Brock, A., Smith, S., De, S., Pascanu, R., Piot, B., and Valko, M. BYOL works even without batch statistics. *arXiv preprint arXiv:2010.10241*, 2020.

Schmidhuber, J., Eldracher, M., and Foltin, B. Semilinear Predictability Minimization Produces Well-Known Feature Detectors. *Neural Computation*, 8(4):773–786, May 1996. ISSN 0899-7667. doi: 10.1162/neco.1996.8.4.773. Conference Name: Neural Computation.

Schwartz, O. and Simoncelli, E. P. Natural signal statistics and sensory gain control. *Nature Neuroscience*, 4(8):819–825, August 2001. ISSN 1546-1726. doi: 10.1038/90526. Number: 8 Publisher: Nature Publishing Group.

Tian, Y., Yu, L., Chen, X., and Ganguli, S. Understanding Self-supervised Learning with Dual Deep Networks. *arXiv preprint arXiv:2010.00578*, 2020.

Tian, Y., Chen, X., and Ganguli, S. Understanding self-supervised Learning Dynamics without Contrastive Pairs. *arXiv:2102.06810 [cs]*, February 2021. arXiv: 2102.06810.

Tishby, N. and Zaslavsky, N. Deep Learning and the Information Bottleneck Principle. *arXiv preprint arXiv:1503.02406*, 2015.

Tishby, N., Pereira, F. C., and Bialek, W. The information bottleneck method. *arXiv preprint arXiv:physics/0004057*, 2000.

Van Horn, G., Mac Aodha, O., Song, Y., Cui, Y., Sun, C., Shepard, A., Adam, H., Perona, P., and Belongie, S. The inaturalist species classification and detection dataset. In *Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

Wang, T. and Isola, P. Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere. *arXiv preprint arXiv:2005.10242*, 2020.

Wu, Y., Kirillov, A., Massa, F., Lo, W.-Y., and Girshick, R. Detectron2. https://github.com/facebookresearch/detectron2, 2019.

You, Y., Gitman, I., and Ginsburg, B. Large batch training of convolutional networks. *arXiv preprint arXiv:1708.03888*, 2017.

Zemel, R. and Hinton, G. E. Discovering viewpoint-invariant relationships that characterize objects. In *NeurIPS*, 1990.

Zhai, X., Oliver, A., Kolesnikov, A., and Beyer, L. S4l: Self-supervised semi-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 1476–1485, 2019.

Zhou, B., Lapedriza, A., Xiao, J., Torralba, A., and Oliva, A. Learning deep features for scene recognition using places database. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.

Ocko, S., Lindsey, J., Ganguli, S., and Deny, S. The emer- gence of multiple retinal cell types through efficient cod- ing of natural movies. In NeurIPS, 2018.

Oord, A. v. d., Li, Y., and Vinyals, O. Representation learn- ing with contrastive predictive coding. arXiv preprint arXiv:1807.03748, 2018.

Redlich, A. N. Redundancy Reduction as a Strategy for Unsupervised Learning. Neural Computation, 5(2):289–304, March 1993a. ISSN 0899-7667. doi: 10.1162/neco.1993.5.2.289. Conference Name: Neural Computation.

Redlich, A. N. Supervised Factorial Learning. Neural Computation, 5(5):750–766, September 1993b. ISSN 0899-7667,1530-888X. doi: 10.1162/neco.1993.5.5.750.

Ren, S., He, K., Girshick, R., and Sun, J. Faster r-cnn: Towards real-time object detection with region proposal networks. In Advances in Neural Information Processing Systems (NeurIPS), 2015. Richemond, P. H., Grill, J.-B., Altch´e, F., Tallec, C., Strub, F., Brock, A., Smith, S., De, S., Pascanu, R., Piot, B., and Valko, M. BYOL works even without batch statistics. arXiv preprint arXiv:2010.10241, 2020. Schmidhuber, J., Eldracher, M., and Foltin, B. Semilinear Predictability Minimization Produces Well-Known Fea- ture Detectors. Neural Computation, 8(4):773–786, May 1996. ISSN 0899-7667. doi: 10.1162/neco.1996.8.4.773. Conference Name: Neural Computation. Schwartz, O. and Simoncelli, E. P. Natural signal statistics and sensory gain control. Nature Neuroscience, 4(8):819–825, August 2001. ISSN 1546-1726. doi: 10.1038/90526. Number: 8 Publisher: Nature Publishing Group. Tian, Y., Yu, L., Chen, X., and Ganguli, S. Understand- ing Self-supervised Learning with Dual Deep Networks. arXiv preprint arXiv:2010.00578, 2020. Tian, Y., Chen, X., and Ganguli, S. Understanding self-supervised Learning Dynamics without Contrastive Pairs. arXiv:2102.*06810 [cs]*, February 2021. arXiv: 2102.06810. Tishby, N. and Zaslavsky, N. Deep Learning and the Information Bottleneck Principle. arXiv preprint arXiv:1503.02406, 2015. Tishby, N., Pereira, F. C., and Bialek, W. The information bottleneck method. arXiv preprint arXiv:physics/0004057, 2000. Van Horn, G., Mac Aodha, O., Song, Y., Cui, Y., Sun, C., Shepard, A., Adam, H., Perona, P., and Belongie, S. The inaturalist species classification and detection dataset. In Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR), 2018. Wang, T. and Isola, P. Understanding Contrastive Repre- sentation Learning through Alignment and Uniformity on the Hypersphere. arXiv preprint arXiv:2005.10242, 2020. Wu, Y., Kirillov, A., Massa, F., Lo, W.-Y., and Gir- shick, R. Detectron2. https://github.com/ facebookresearch/detectron2, 2019. You, Y., Gitman, I., and Ginsburg, B. Large batch training of convolutional networks. arXiv preprint arXiv:1708.03888, 2017. Zemel, R. and Hinton, G. E. Discovering viewpoint- invariant relationships that characterize objects. In NeurIPS, 1990. Zhai, X., Oliver, A., Kolesnikov, A., and Beyer, L. S4l: Self- supervised semi-supervised learning. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 1476–1485, 2019. Zhou, B., Lapedriza, A., Xiao, J., Torralba, A., and Oliva, A. Learning deep features for scene recognition using places database. In Advances i n Neural Information Processing Systems (NeurIPS), 2014.

A. Connection between BARLOW TWINS and the Information Bottleneck Principle

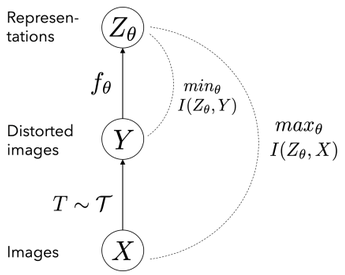


Figure 6. The information bottleneck principle applied to self-supervised learning (SSL) posits that the objective of SSL is to learn a representation Z_θ which is informative about the image sample, but invariant (i.e. uninformative) to the specific distortions that are applied to this sample. BARLOW TWINS can be viewed as a specific instanciation of the information bottleneck objective.

We explore in this appendix the connection between BARLOW TWINS' loss function and the *Information Bottleneck* (IB) principle (Tishby & Zaslavsky, 2015; Tishby et al., 2000).

As a reminder, BARLOW TWINS' loss function is given by:

$$\mathcal{L}_{\mathcal{BT}} \triangleq \underbrace{\sum_i (1 - \mathcal{C}_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} \mathcal{C}_{ij}{}^2}_{\text{redundancy reduction term}} \quad (3)$$

where λ is a positive constant trading off the importance of the first and second terms of the loss, and where $\mathcal{C}$ is the cross-correlation matrix computed between the outputs of the two identical networks along the batch dimension :

$$\mathcal{C}_{ij} \triangleq \frac{\sum_b z^A_{b,i} z^B_{b,j}}{\sqrt{\sum_b (z^A_{b,i})^2} \sqrt{\sum_b (z^B_{b,j})^2}} \quad (4)$$

where b indexes batch samples and i, j index the vector dimension of the networks' outputs. $\mathcal{C}$ is a square matrix with size the dimensionality of the network's output, and with values comprised between -1 (i.e. perfect anti-correlation) and 1 (i.e. perfect correlation).

Applied to self-supervised learning, the IB principle posits that a desirable representation should be as informative as possible about the sample represented while being as invariant (i.e. non-informative) as possible to distortions of that sample (here the data augmentations used) (Fig. 6). This trade-off is captured by the following loss function:

$$\mathcal{IB}_\theta \triangleq I(Z_\theta, Y) - \beta I(Z_\theta, X) \quad (5)$$

where $I(.,.)$ denotes mutual information and β is a positive scalar trading off the desideratas of preserving information and being invariant to distortions.

Using a classical identity for mutual information, we can rewrite equation 5 as:

$$\mathcal{IB}_\theta = [H(Z_\theta) - \cancelto{0}{H(Z_\theta|Y)}] - \beta [H(Z_\theta) - H(Z_\theta|X)] \quad (6)$$

where $H(.)$ denotes entropy. The conditional entropy $H(Z_\theta|Y)$ —the entropy of the representation conditioned on a specific distorted sample— cancels to 0 because the function f_θ is deterministic, and so the representation Z_θ conditioned on the input sample Y is perfectly known and has zero entropy. Since the overall scaling factor of the loss function is not important, we can rearrange equation 6 as:

$$\mathcal{IB}_\theta = H(Z_\theta|X) + \frac{1-\beta}{\beta} H(Z_\theta) \quad (7)$$

Measuring the entropy of a high-dimensional signal generally requires vast amounts of data, much larger than the size of a single batch. In order to circumvent this difficulty, we make the simplifying assumption that the representation Z is distributed as a Gaussian. The entropy of a Gaussian distribution is simply given by the logarithm of the determinant of its covariance function (up to a constant corresponding to the assumed discretization level that we ignore here) (Cai et al., 2015). The loss function becomes:

$$\mathcal{IB}_\theta = \mathbb{E}_X log\, |\mathcal{C}_{Z_\theta|X}| + \frac{1-\beta}{\beta} log\, |\mathcal{C}_{Z_\theta}| \quad (8)$$

This equation is still not exactly the one we optimize for in practice (see eqn. 3 and 4). Indeed, our loss function is only connected to the IB loss given by eqn. 8 through the following simplifications and approximations:

- In the case where $\beta <= 1$, it is easy to see from eqn. 8 that the best solution to the IB trade-off is to set the representation to a constant that does not depend on the input. This trade-off thus does not lead to interesting representations and can be ignored. When $\beta > 1$, we note that the second term of eqn. 8 is preceded by a negative constant. We can thus simply replace $\frac{1-\beta}{\beta}$ by a new positive constant λ, preceded by a negative sign.

A. BARLOW TWINS 与信息瓶颈原则的联系

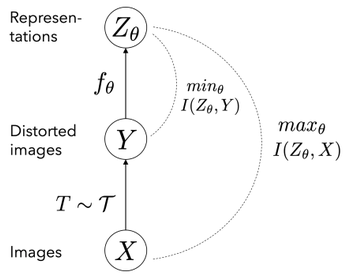


图6. 用于自监督学习（SSL）的信息瓶颈原则认为，SSL 应学习包含图像样本信息、但对样本的具体扰动保持不变（即不包含扰动信息）的表征 Z_θ。BARLOW TWINS 可视为信息瓶颈目标的一种具体实例。

本附录探讨 BARLOW TWINS 损失与信息瓶颈（IB）原则的联系（Tishby & Zaslavsky, 2015; Tishby et al., 2000）。

回顾一下，BARLOW TWINS 损失为：

$$\mathcal{L}_{\mathcal{BT}} \triangleq \underbrace{\sum_i (1 - \mathcal{C}_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} \mathcal{C}_{ij}{}^2}_{\text{redundancy reduction term}} \quad (3)$$

其中 λ 是权衡损失第一项与第二项重要性的正常数；$\mathcal{C}$ 是沿批量维度计算的两个相同网络输出之间的交叉相关矩阵：

$$\mathcal{C}_{ij} \triangleq \frac{\sum_b z^A_{b,i} z^B_{b,j}}{\sqrt{\sum_b (z^A_{b,i})^2} \sqrt{\sum_b (z^B_{b,j})^2}} \quad (4)$$

其中 b 索引批次样本，i、j 索引网络输出向量的维度。$\mathcal{C}$ 是边长等于网络输出维度的方阵，其值在−1（完全负相关）与1（完全正相关）之间。

在自监督学习中，IB 原则要求理想表征尽可能包含所表示样本的信息，同时尽可能对扰动保持不变（即不包含扰动信息）

该样本的扰动（此处为数据增强）（图6）。这种权衡由以下损失表达：

$$\mathcal{IB}_\theta \triangleq I(Z_\theta, Y) - \beta I(Z_\theta, X) \quad (5)$$

其中 $I(.,.)$ 表示互信息；β 是正常数，权衡保留信息和对扰动保持不变这两个要求。

用互信息的经典恒等式，可将式5改写为：

$$\mathcal{IB}_\theta = [H(Z_\theta) - \cancelto{0}{H(Z_\theta|Y)}] - \beta [H(Z_\theta) - H(Z_\theta|X)] \quad (6)$$

其中 $H(.)$ 表示熵。条件熵 $H(Z_\theta|Y)$，即给定某个扰动样本时表征的熵，等于0，因为函数 f_θ 是确定性的；给定输入 Y，表征 Z_θ 完全确定，因而熵为零。损失的整体缩放系数并不重要，因此可将式6改写为：

$$\mathcal{IB}_\theta = H(Z_\theta|X) + \frac{1-\beta}{\beta} H(Z_\theta) \quad (7)$$

估计高维信号的熵通常需要远超单个批次的大量数据。为绕过这一困难，我们作简化假设：表征 Z 服从高斯分布。高斯分布的熵由其协方差行列式的对数给出，差一个与假定离散化精度有关的常数，此处忽略（Cai et al., 2015）。损失函数变为：

$$\mathcal{IB}_\theta = \mathbb{E}_X log\, |\mathcal{C}_{Z_\theta|X}| + \frac{1-\beta}{\beta} log\, |\mathcal{C}_{Z_\theta}| \quad (8)$$

该式仍不完全等于我们实际优化的目标（见式3和4）。事实上，我们的损失仅通过以下简化和近似与式8的 IB 损失相联系：

- 当 $\beta <= 1$ 时，由式8容易看出，IB 权衡的最佳解是与输入无关的常量表征。这种权衡无法得到有意义的表征，可以忽略。当 $\beta > 1$ 时，式8第二项前的系数为负。因此，可将 $\frac{1-\beta}{\beta}$ 替换为一个新的正常数 λ，并在前面加负号。

- In practice, we find that directly optimizing the determinant of the covariance matrices does not lead to SoTA solutions. Instead, we replace the second term of the loss in eqn. 8 (maximizing the information about samples), by the proxy which consist in simply minimizing the Frobenius norm of the cross-correlation matrix. If the representations are assumed to be re-scaled to 1 along the batch dimension before entering the loss (an assumption we are free to make since the cross-correlation matrix is invariant to this re-scaling), this minimization only affects the off-diagonal terms of the covariance matrix (the diagonal terms being fixed to 1 by the re-scaling) and encourages them to be as close to 0 as possible. It is clear that this surrogate objective, which consists in decorrelating all output units, has the same global optimum than the original information maximization objective.

- For consistency with eqn. 8, the second term in BARLOW TWINS' loss should be computed from the auto-correlation matrix of one of the twin networks, instead of the cross-correlation matrix between twin networks. In practice, we do not see a strong difference in performance between these alternatives.

- Similarly, it can easily be shown that the first term of eqn. 8 (minimizing the information the representation contains about the distortions) has the same global optimum than the first term of eqn. 3, which maximizes the alignment between representations of pairs of distorted samples.

B. Evaluations on ImageNet

B.1. Linear evaluation on ImageNet

The linear classifier is trained for 100 epochs with a learning rate of 0.3 and a cosine learning rate schedule. We minimize the cross-entropy loss with the SGD optimizer with momentum and weight decay of 10^{-6}. We use a batch size of 256. At training time we augment an input image by taking a random crop, resizing it to 224×224, and optionally flipping the image horizontally. At test time we resize the image to 256×256 and center-crop it to a size of 224×224.

B.2. Semi-supervised training on ImageNet

We train for 20 epochs with a learning rate of 0.002 for the ResNet-50 and 0.5 for the final classification layer. The learning rate is multiplied by a factor of 0.2 after the 12th and 16th epoch. We minimize the cross-entropy loss with the SGD optimizer with momentum and do not use weight decay. We use a batch size of 256. The image augmentations are the same as in the linear evaluation setting.

C. Transfer Learning

C.1. Linear evaluation

We follow the exact settings from PIRL (Misra & van der Maaten, 2019) for evaluating linear classifiers on the Places-205, VOC07 and iNaturalist2018 datasets. For Places-205 and iNaturalist2018 we train a linear classifier with SGD (14 epochs on Places-205, 84 epochs on iNaturalist2018) with a learning rate of 0.01 reduced by a factor of 10 at two equally spaced intervals, a weight decay of 5×10^{-4} and SGD momentum of 0.9. We train SVM classifiers on the VOC07 dataset where the C values are computed using cross-validation.

C.2. Object Detection and Instance Segmentation

We use the detectron2 library (Wu et al., 2019) for training the detection models and closely follow the evaluation settings from (He et al., 2019). The backbone ResNet50 network for Faster R-CNN (Ren et al., 2015) and Mask R-CNN (He et al., 2017) is initialized using our BARLOW TWINS pretrained model.

VOC07+12 We use the VOC07+12 `trainval` set of $16K$ images for training a Faster R-CNN (Ren et al., 2015) C-4 backbone for $24K$ iterations using a batch size of 16 across 8 GPUs using SyncBatchNorm. The initial learning rate for the model is 0.1 which is reduced by a factor of 10 after $18K$ and $22K$ iterations. We use linear warmup (Goyal et al., 2017) with a slope of 0.333 for 1000 iterations.

COCO We train Mask R-CNN (He et al., 2017) C-4 backbone on the COCO 2017 `train` split and report results on the `val` split. We use a learning rate of 0.03 and keep the other parameters the same as in the $1\times$ schedule in detectron2.

• 实践中，直接优化协方差矩阵的行列式并不能得到最优水平的解。因此，我们以简单最小化交叉相关矩阵 Frobenius 范数这一替代目标，取代式8的第二项（最大化关于样本的信息）。假设表征在进入损失前沿批量维度缩放到1——交叉相关矩阵对这种缩放不变，因此可以自由作此假设——则最小化只影响协方差矩阵的非对角项，因为缩放使对角项固定为1；它鼓励非对角项尽可能接近0。显然，这种使全部输出单元去相关的替代目标，与原信息最大化目标具有相同的全局最优解。

• 为与式8一致，BARLOW TWINS 损失第二项应从一个网络的自相关矩阵计算，而非双网络间的交叉相关矩阵。实际中，两种方案的性能差异并不明显。

• 同样容易证明，式8第一项（最小化表征所含的扰动信息）与式3第一项（最大化扰动样本对表征的对齐程度）具有相同的全局最优解。

B. ImageNet 评估

B.1. ImageNet 线性评估

线性分类器训练100轮，学习率0.3，采用余弦调度。使用带动量的 SGD 最小化交叉熵，权重衰减为 10^{-6}，批量大小256。训练时随机裁剪输入图像，缩放至 224×224，并可随机水平翻转。测试时将图像缩放至 256×256，再中心裁剪到 224×224。

B.2. ImageNet 半监督训练

我们训练20轮，ResNet-50 学习率0.002，最终分类层学习率0.5。在第12及16轮后，学习率各乘以0.2。用带动量的 SGD 最小化交叉熵，不使用权重衰减，批量大小256。图像增强与线性评估相同。

C. 迁移学习

C.1. 线性评估

我们完全遵循 PIRL（Misra & van der Maaten, 2019）的设置，在 Places-205、VOC07 和 iNaturalist2018 上评估线性分类器。Places-205 和 iNaturalist2018 用 SGD 训练，分别训练14轮和84轮；学习率0.01，在两个等间隔节点各降低10倍；权重衰减 5×10^{-4}，SGD 动量0.9。VOC07 训练 SVM 分类器，C 值通过交叉验证确定。

C.2. 目标检测与实例分割

检测模型使用 detectron2 库（Wu et al., 2019）训练，严格遵循 He et al.（2019）的评估设置。Faster R-CNN（Ren et al., 2015）及 Mask R-CNN（He et al., 2017）的 ResNet50 主干以 BARLOW TWINS 预训练模型初始化。

VOC**07+12** 我们使用 VOC07+12 的 trainval 集，共 $16K$ 张图像，以 Faster R-CNN（Ren et al., 2015）的 C-4 主干训练 $24K$ 次迭代；8 块 GPU，批量 16，使用 SyncBatchNorm。初始学习率0.1，在 $18K$ 和 $22K$ 次迭代后各降低10倍。前1000次迭代使用斜率0.333的线性预热（Goyal et al., 2017）。

COCO 我们在 COCO 2017 train 划分上训练 C-4 主干的 Mask R-CNN（He et al., 2017），在 val 划分上报告结果。学习率0.03，其余参数与 detectron2 的 $1\times$ 调度相同。